

Thread - 2024-12-08

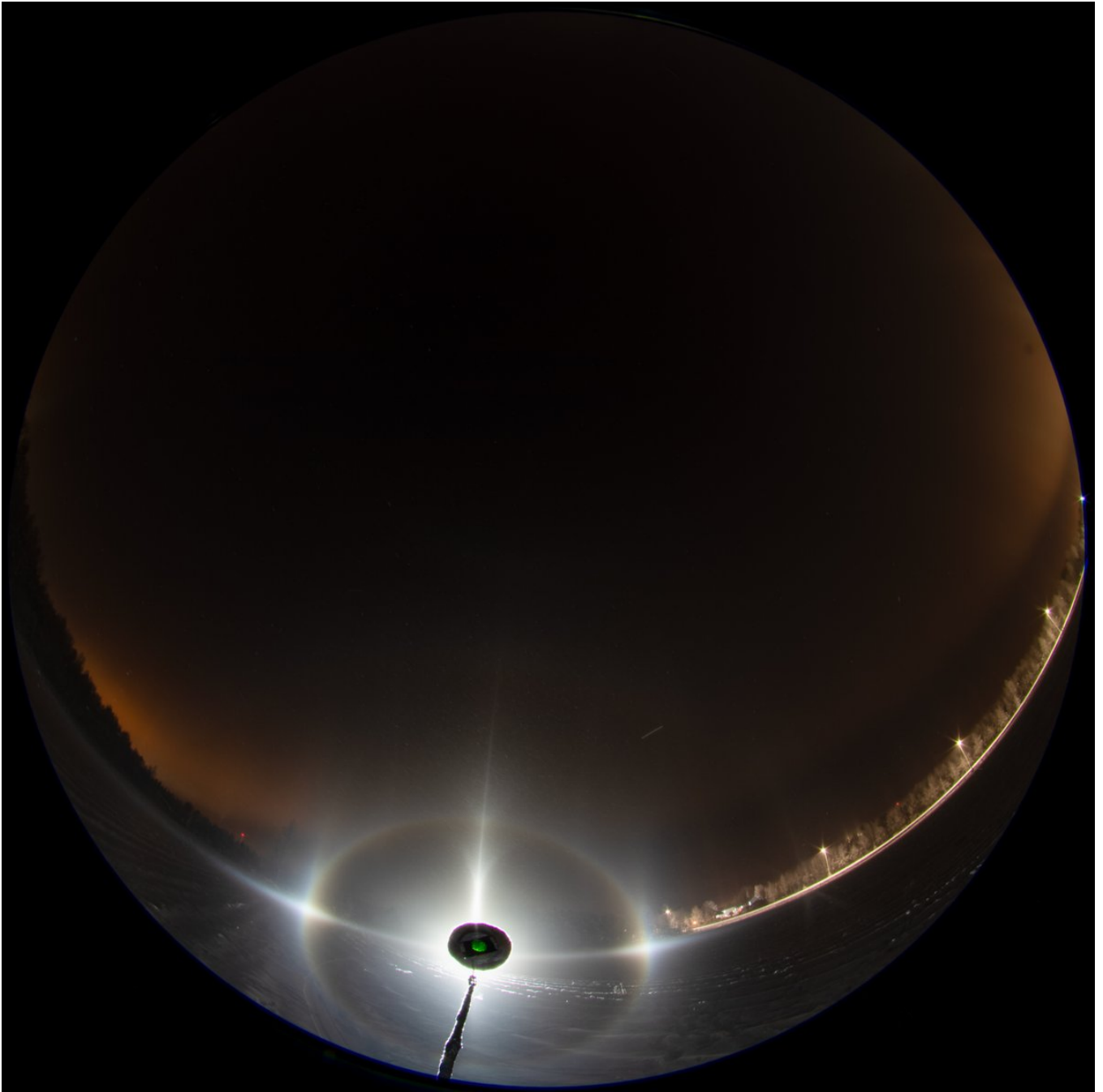
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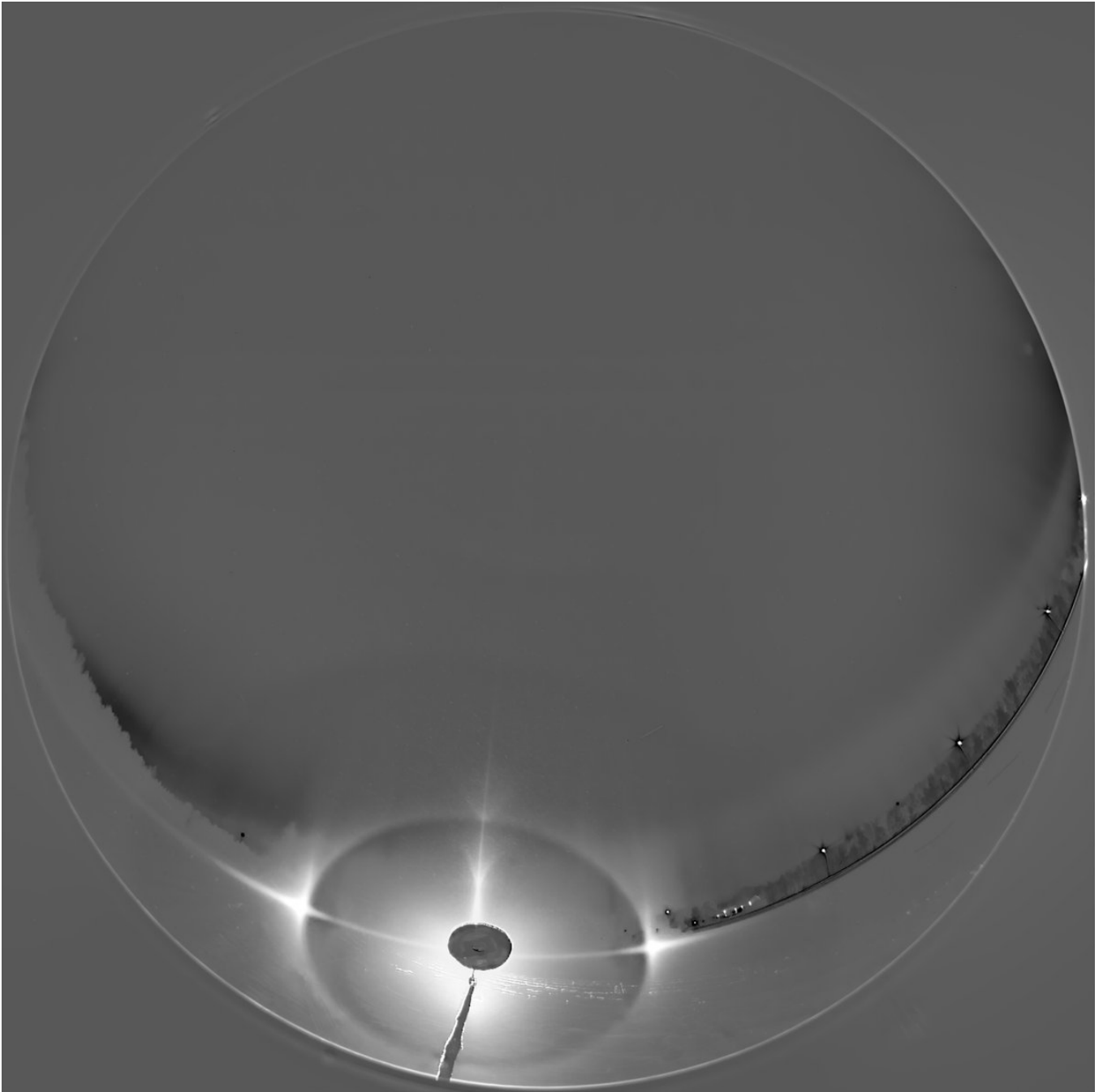
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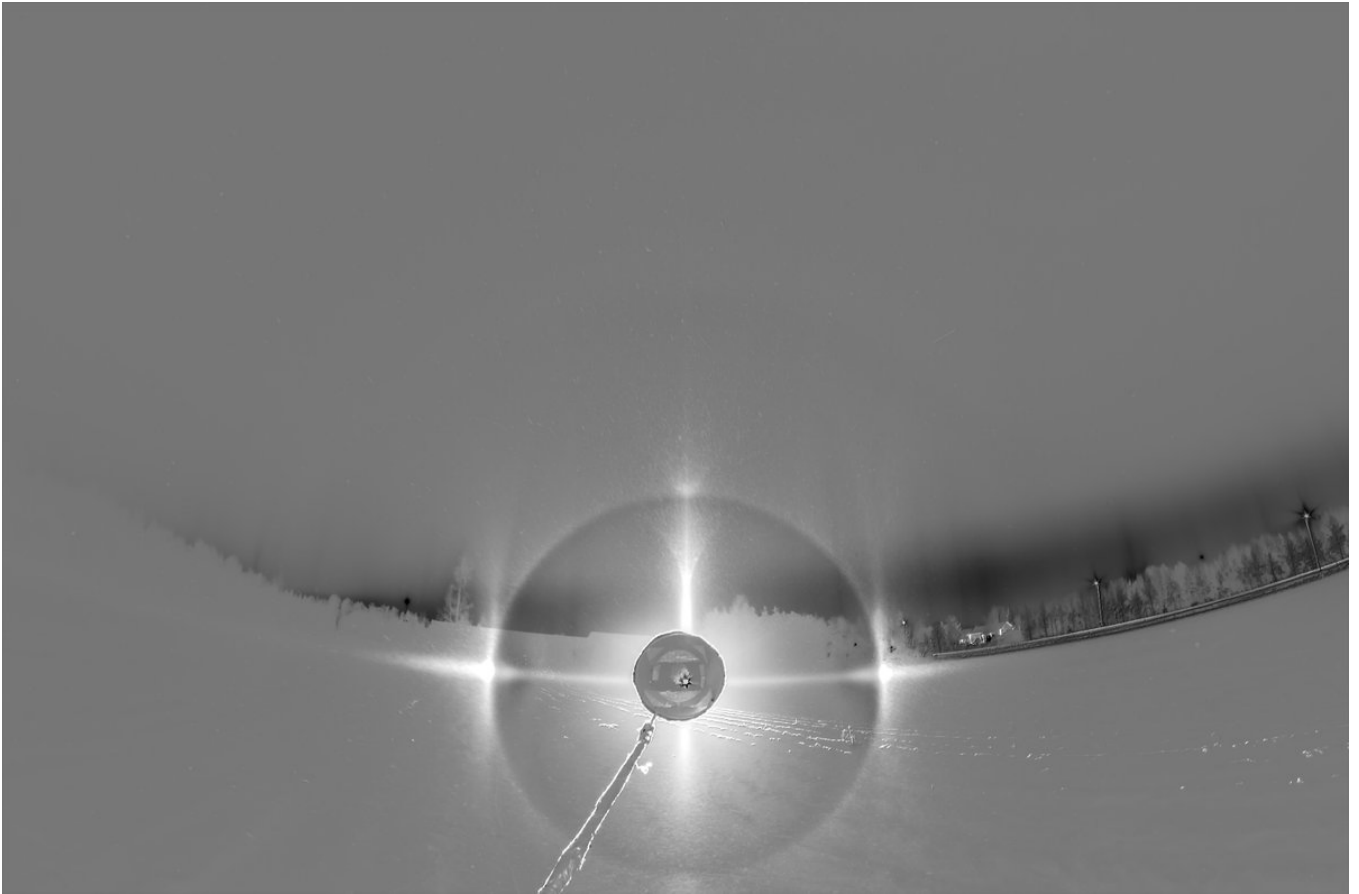
It was an anomaly that last winter gave only one Moilanen arc. Here is that display (11/12 November 2023 Rovaniemi) followed by a profusion of crystal photos. Given the Moilanen arc characteristically has long wings – it is not a stub like 9° plate arc – the crystal must be something elongated. And there are indeed plenty of those elongated crossed plates here that we have learned to associate with Moilanen arc displays. But as always, they look to me a pretty motley bunch. If they indeed contain the wedge that makes the Moilanen arc, I am hard pressed to see how they could orient so as to maintain identical rotation about the long axis.

That second V, on top the 22° halo, is weak but quite sharp, suggesting it is rather a Parry arc. Unfortunately the views are rather tight on the bottom part the 22° halo, but it looks like there is a tangential arc there. Which would mean not a Parry arc after all but tangent arc. But simulation with column oriented crystals having full 360° degree rotation don't look convincing. An intermediary between Parry and tangent arc, from poorly oriented Parry crystals that rotate some 5 degrees, looks better and it also produces a lower arc that touches the 22° halo

Related, there is an interesting observation by Moilanen and me from 2015 where the bottom arc is absent: <https://t.co/qk5WQpDyGy> You need non-regular hexagons to remove the lower arc. Trapezoids to full triangles. And plate shaped because the arc is just a stub (small crystal size may also play a part here). There is evidence of plate shaped triangular Parry crystals from big displays. But could instead 90° degree crossed plates be the explanation here? If there was a trapezoid attached upright in the middle of a more regular crystal and this combo would fall with the trapezoid on the top, only the upper arc would be visible.

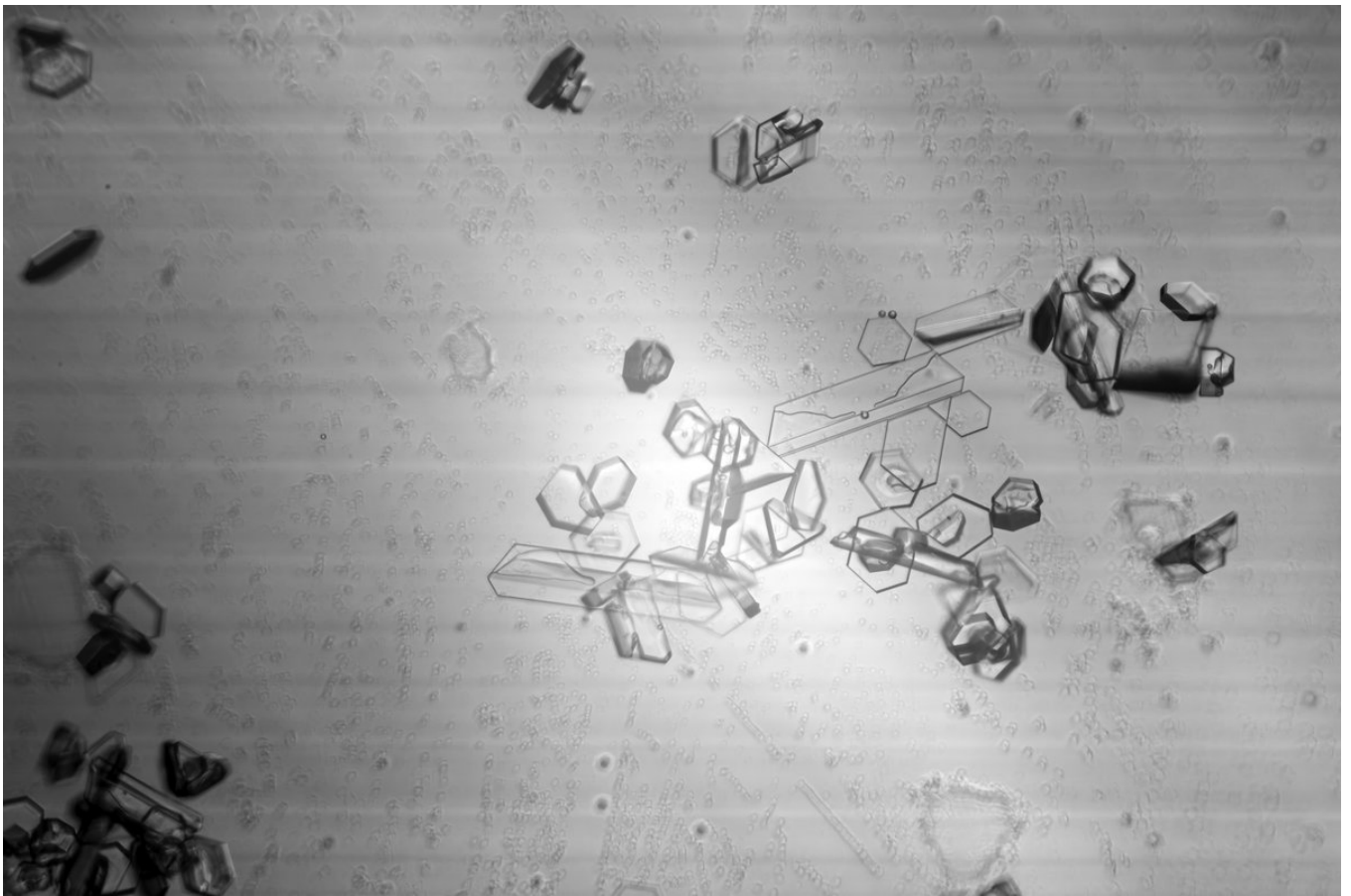
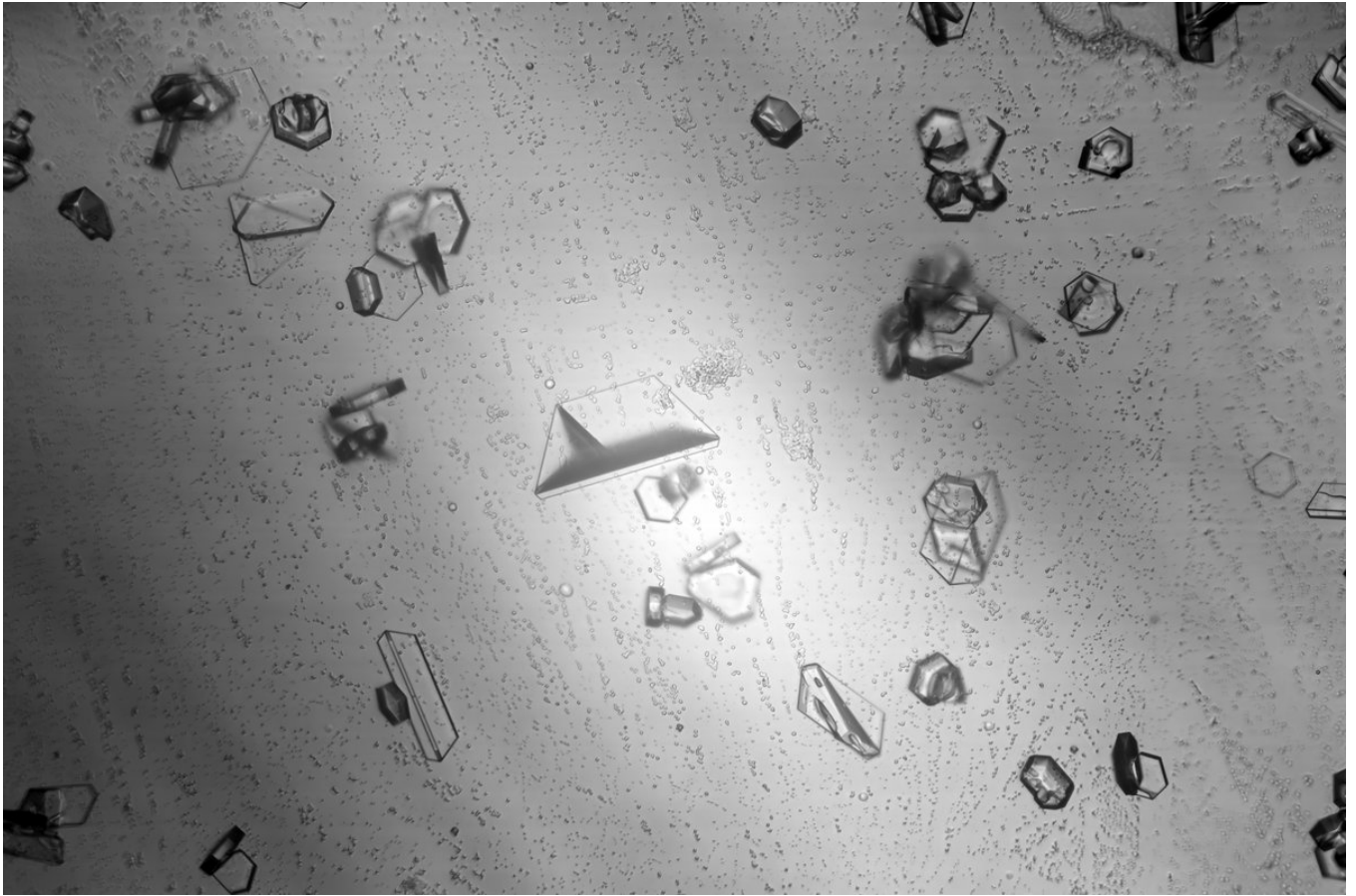


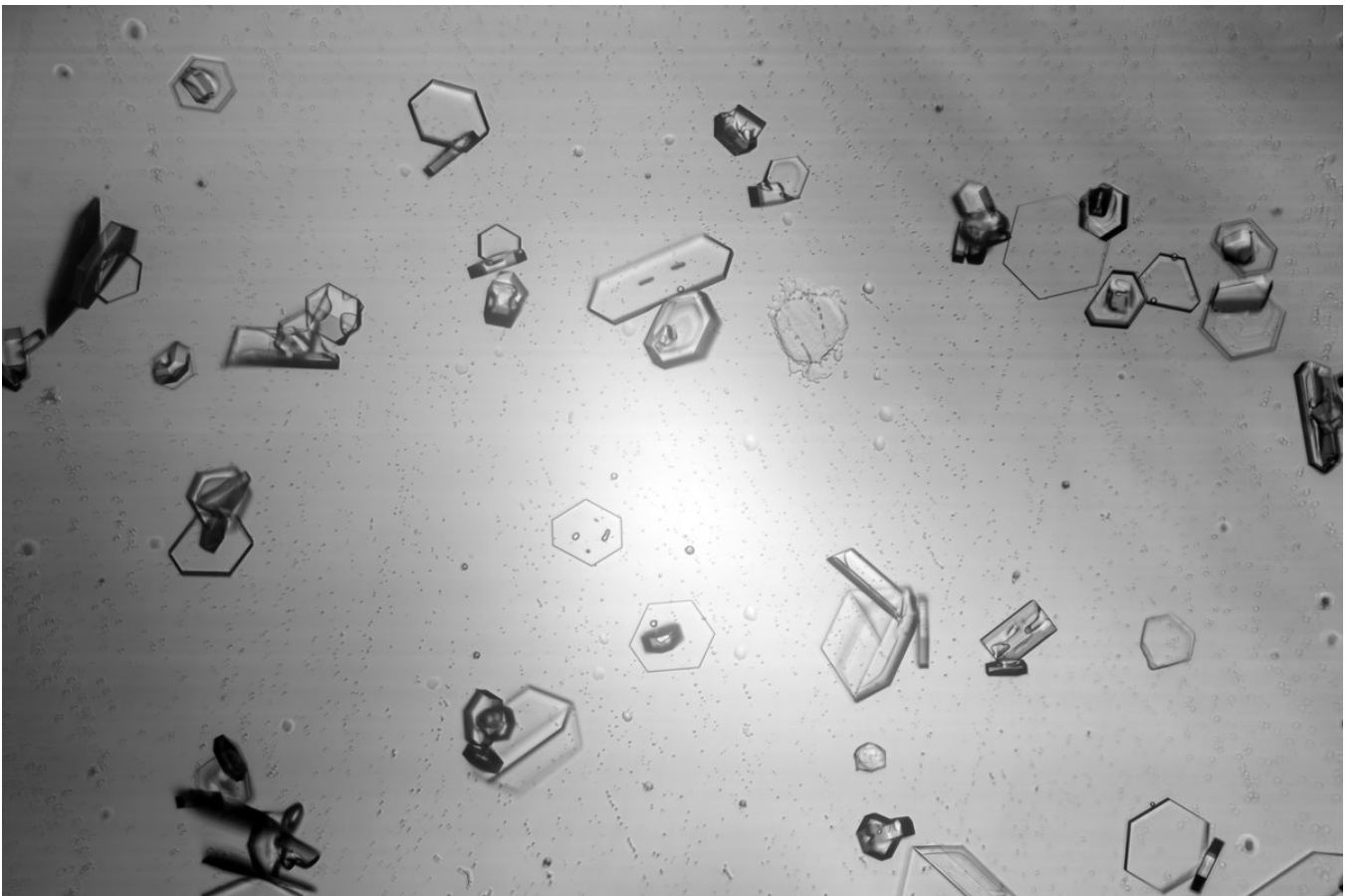
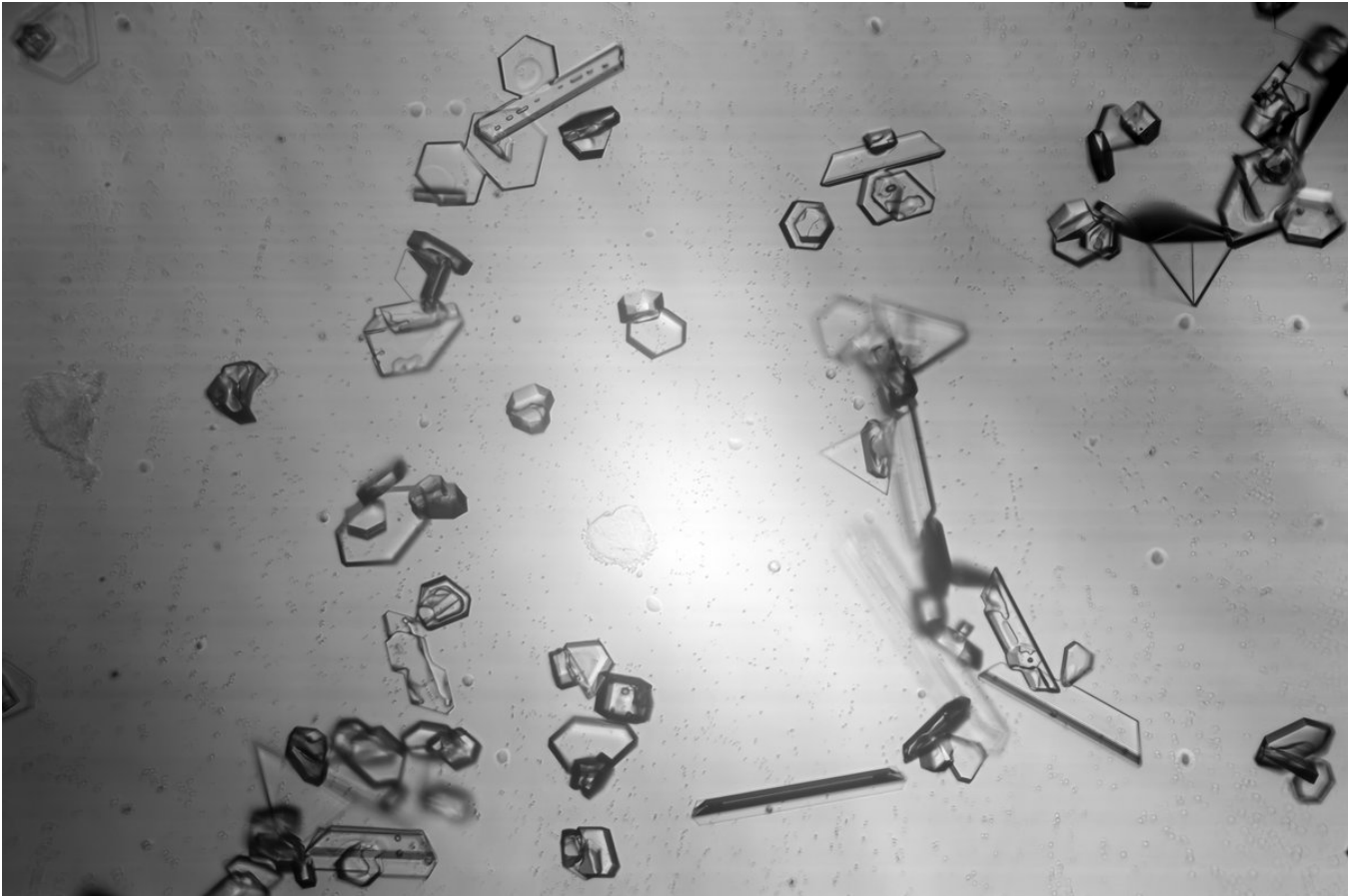




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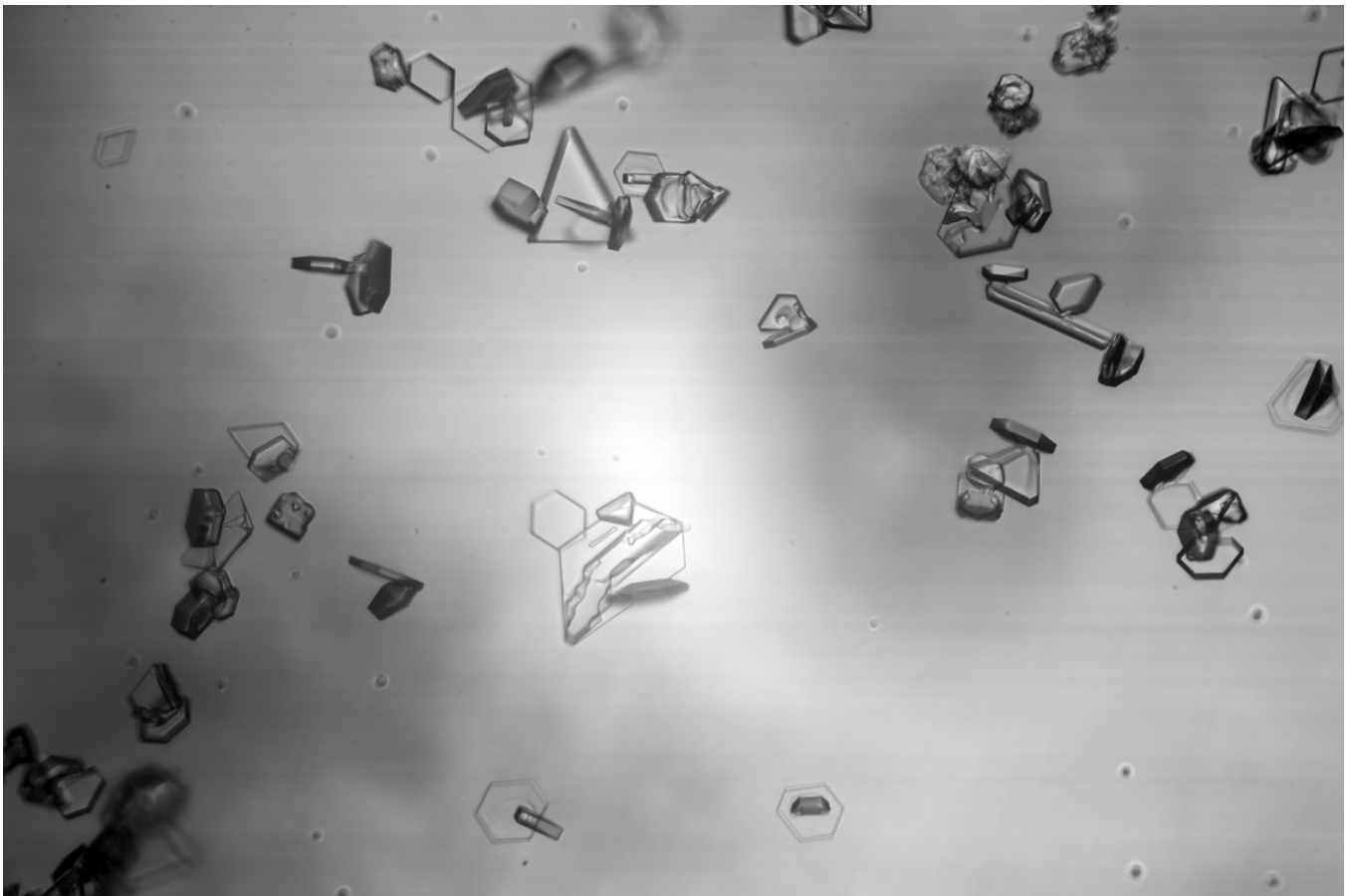
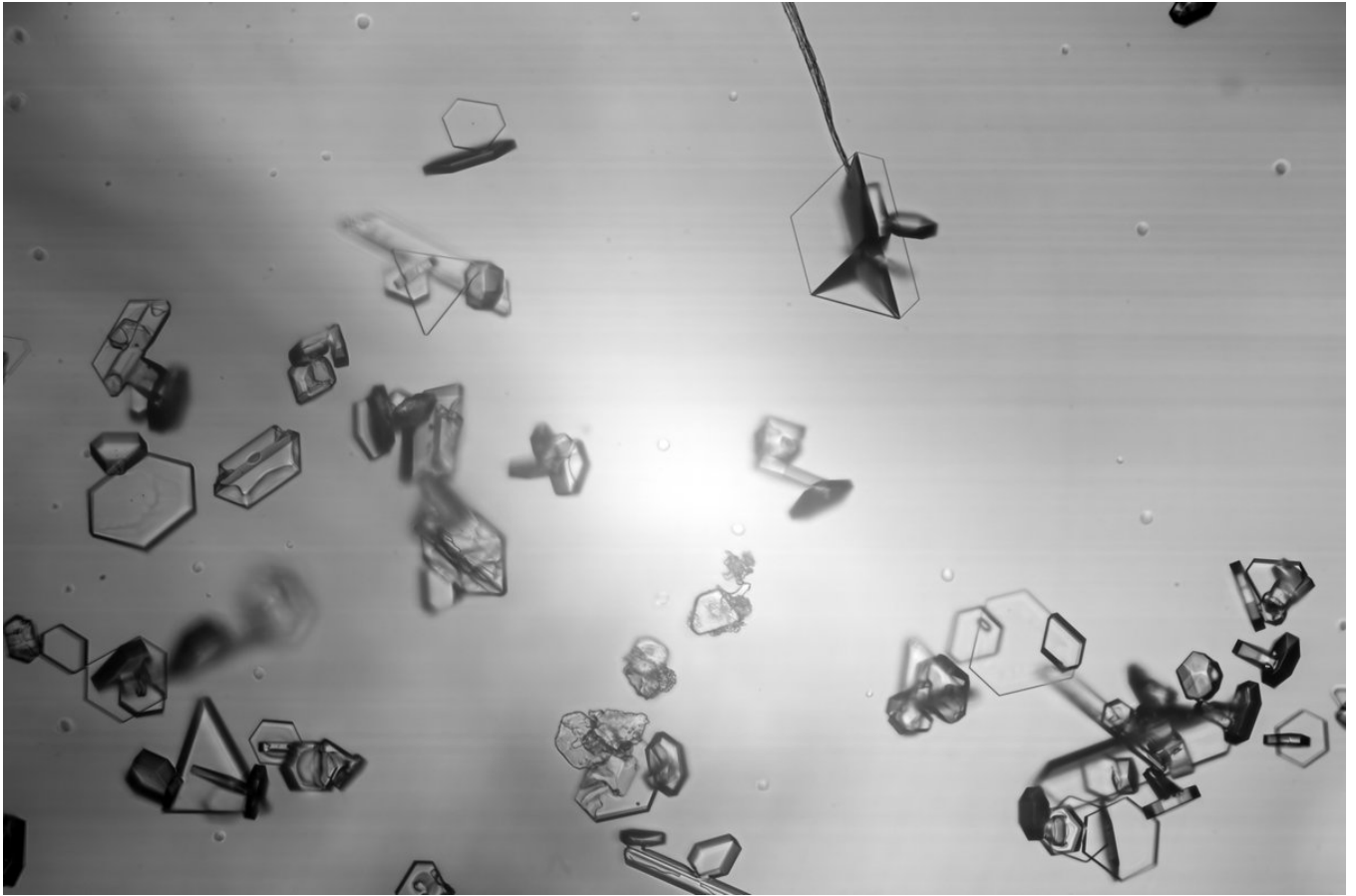
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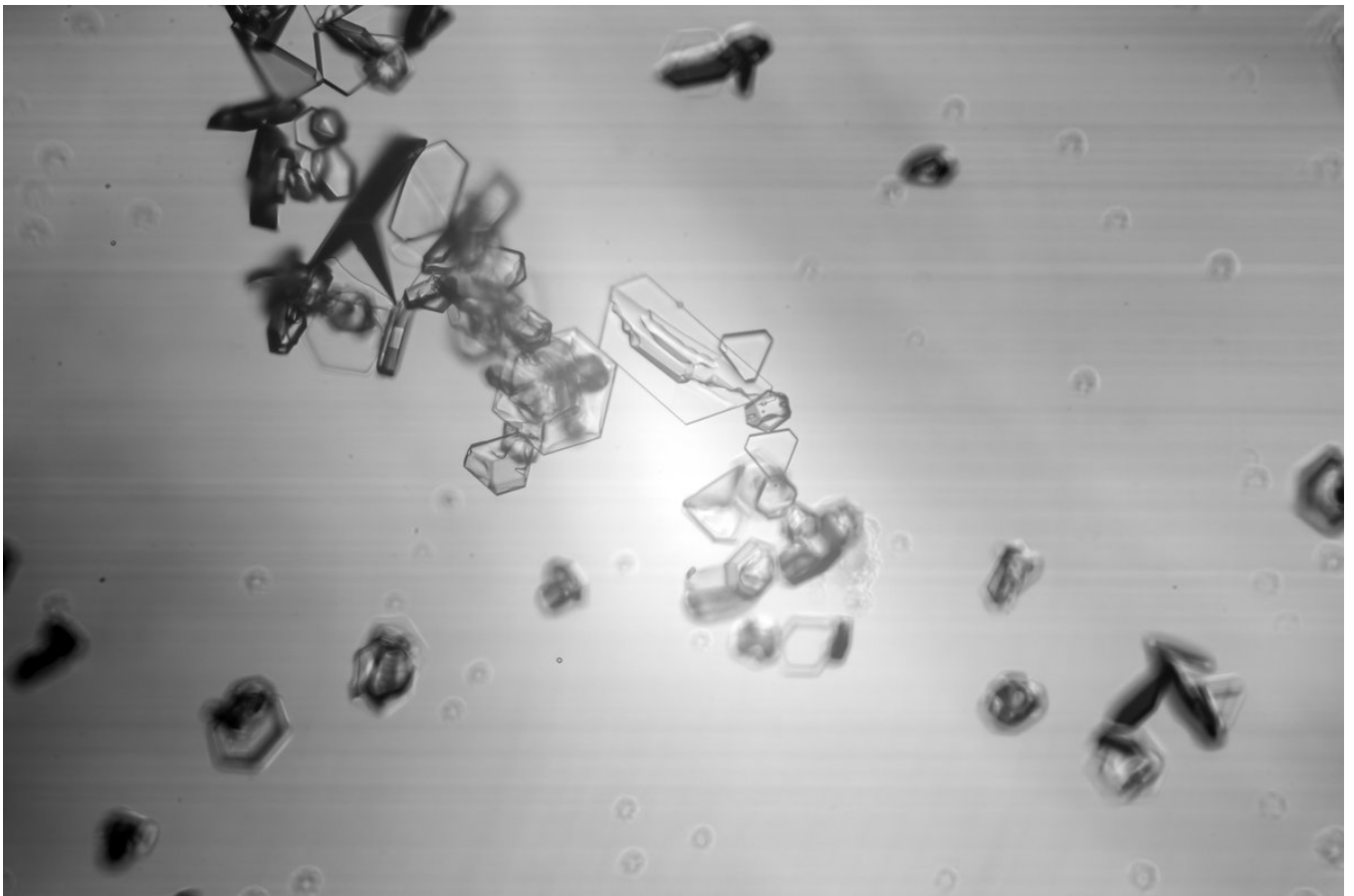
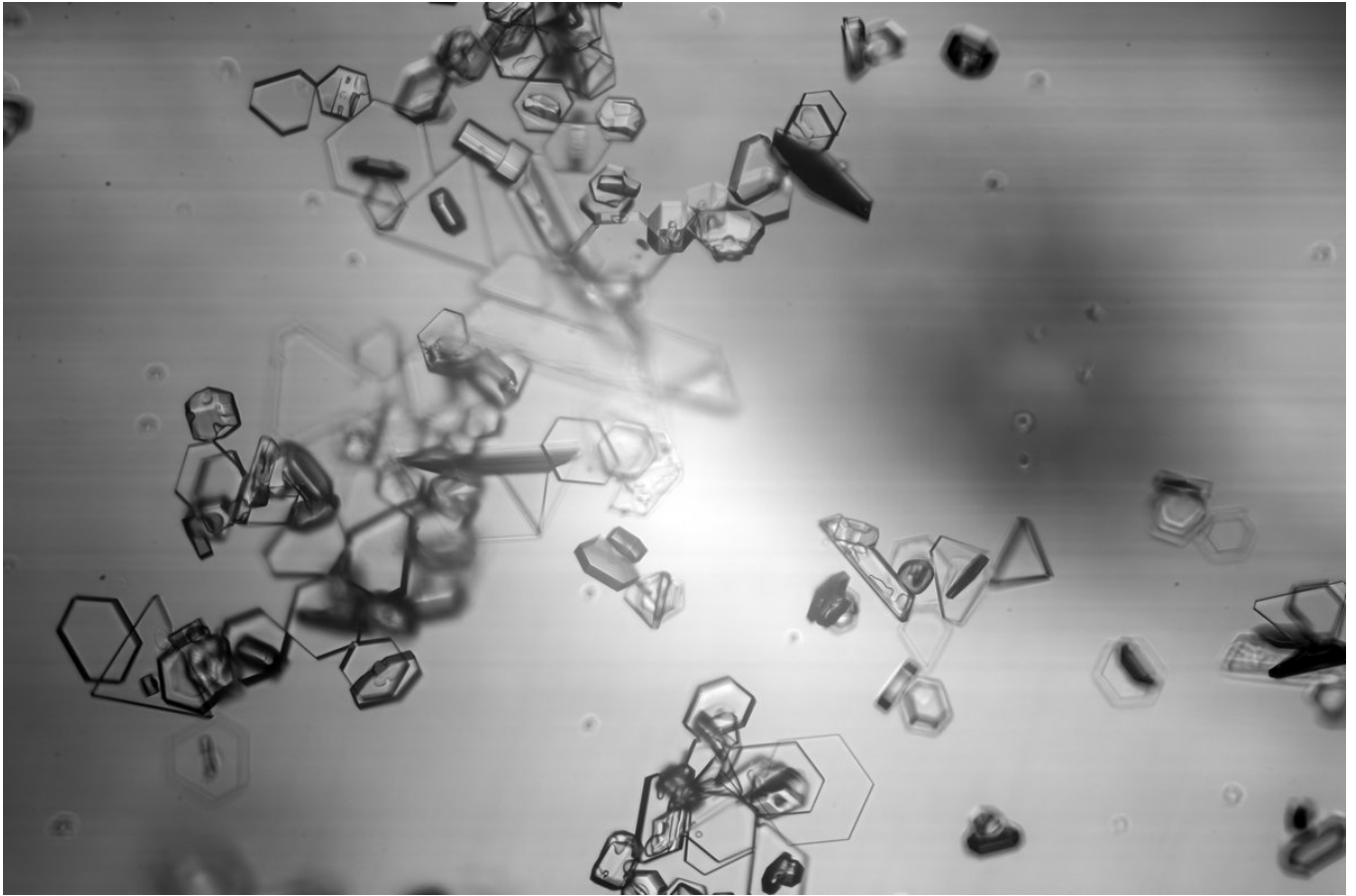




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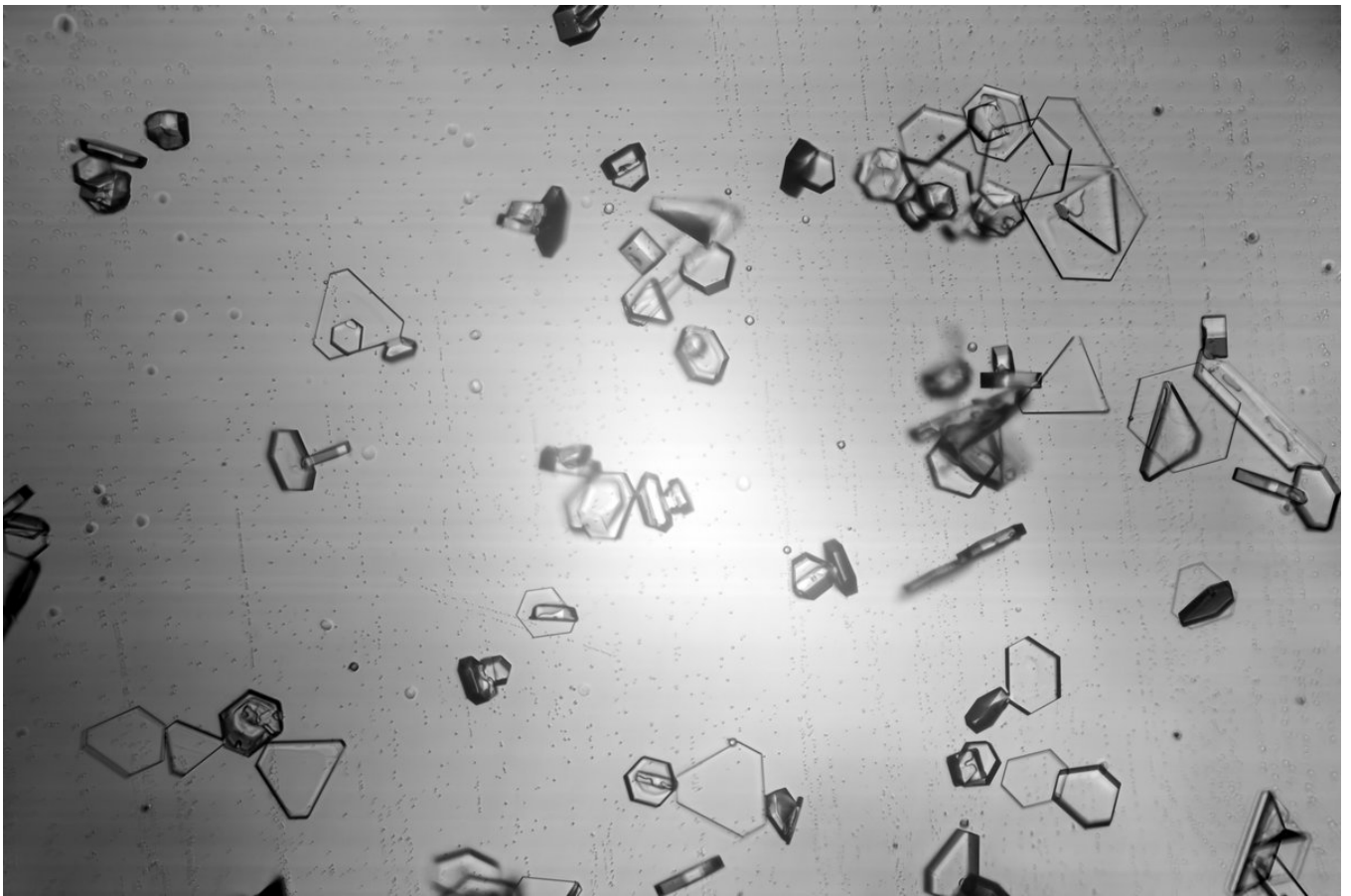
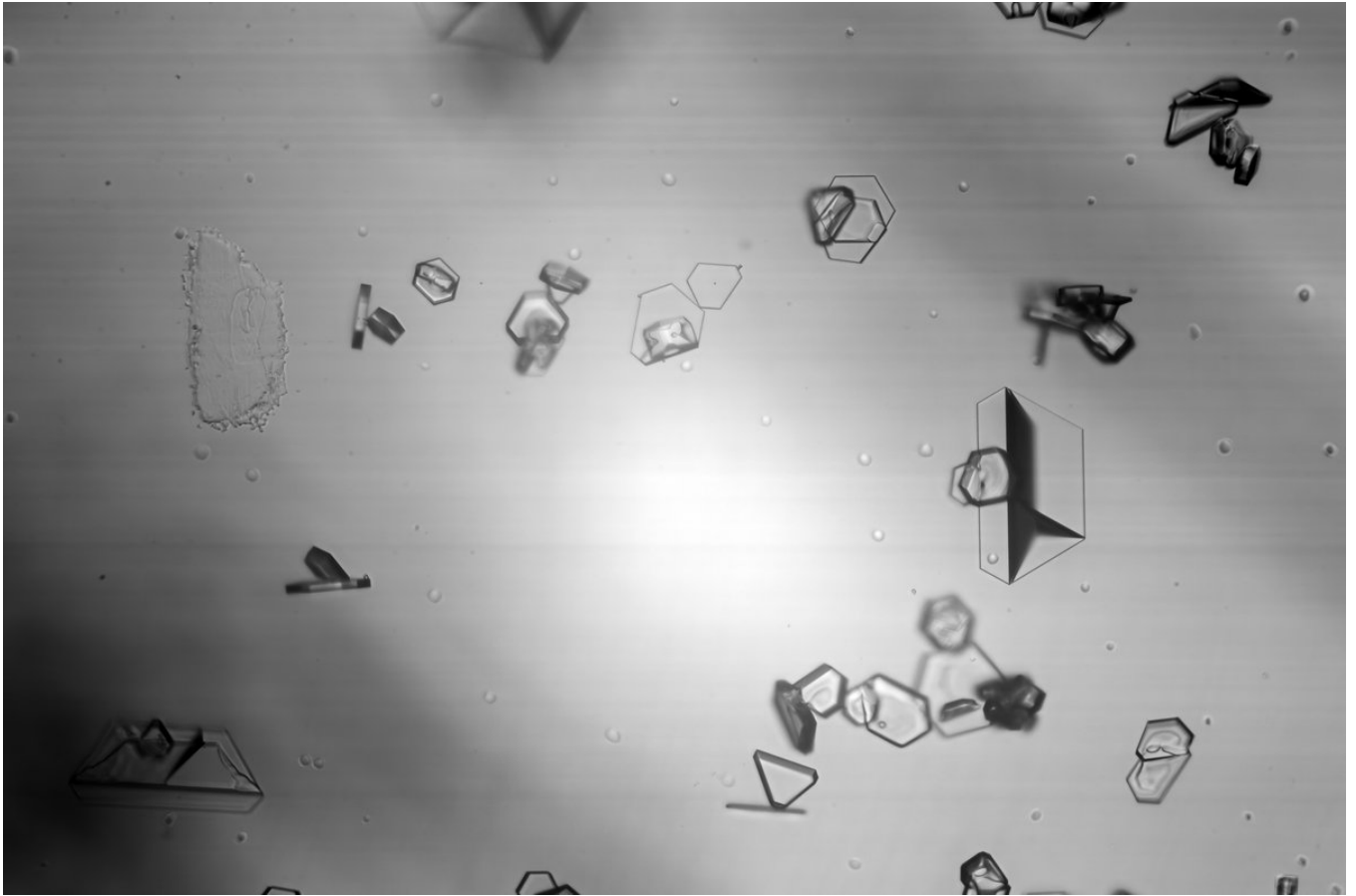
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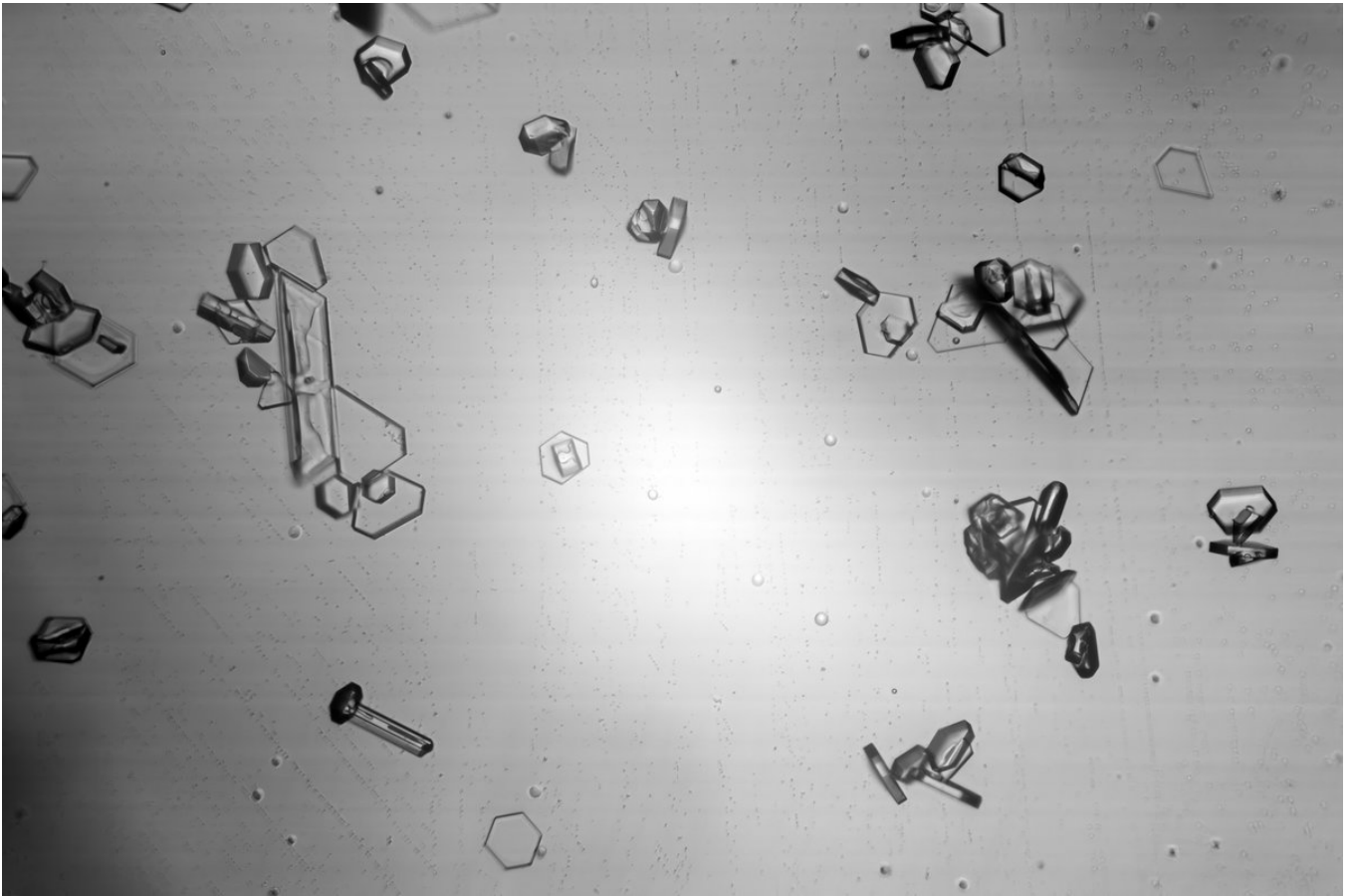
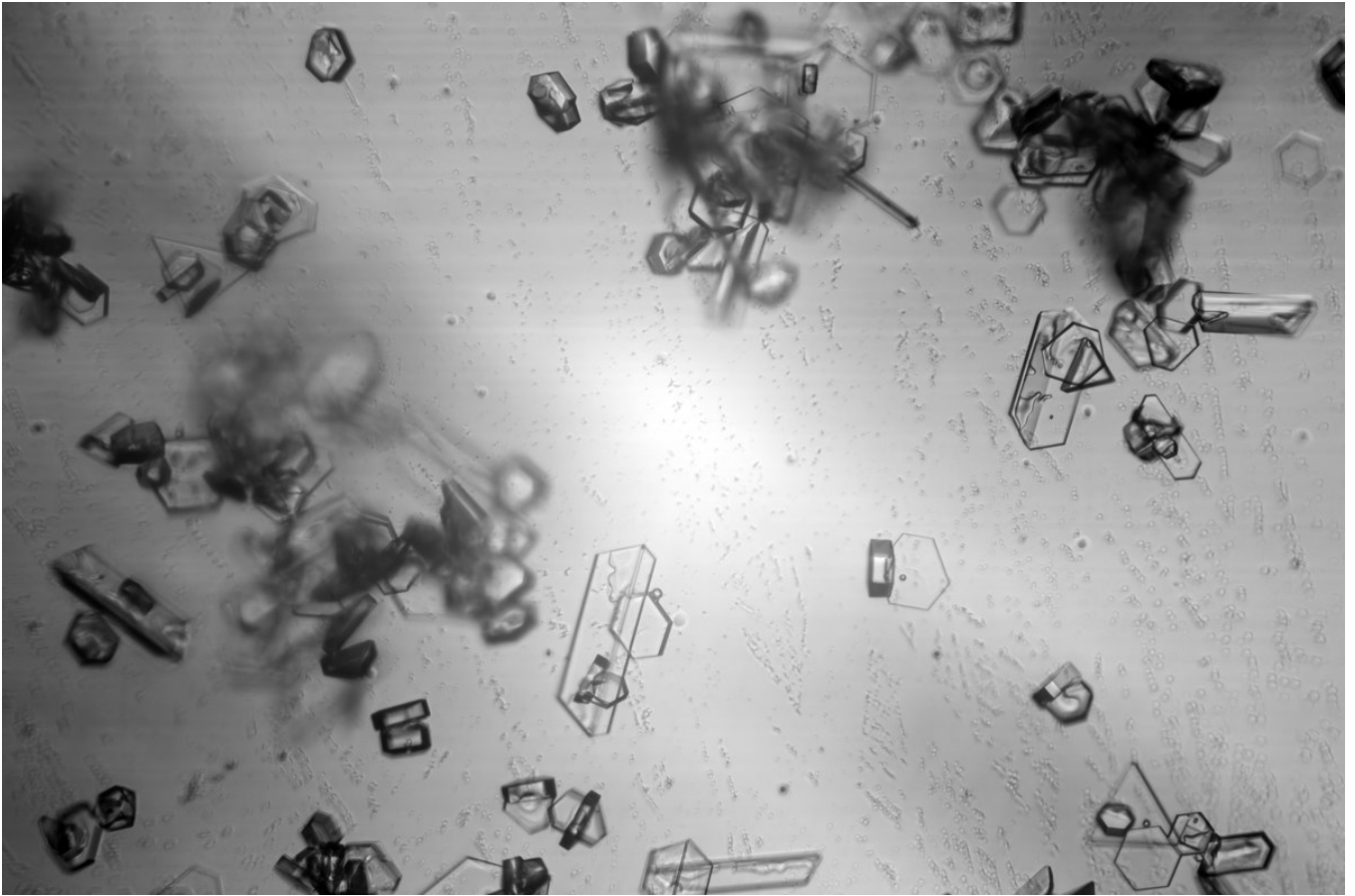




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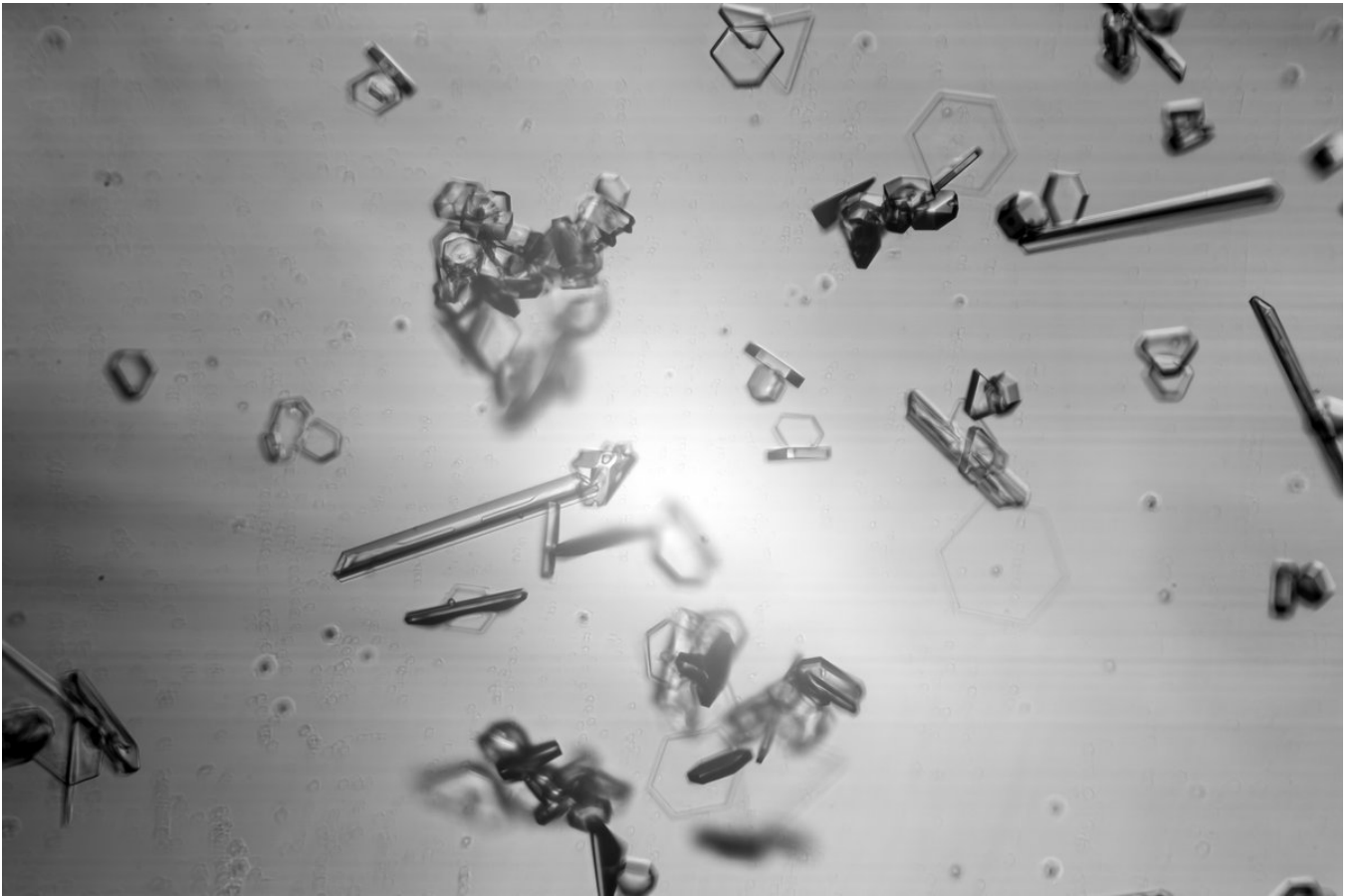
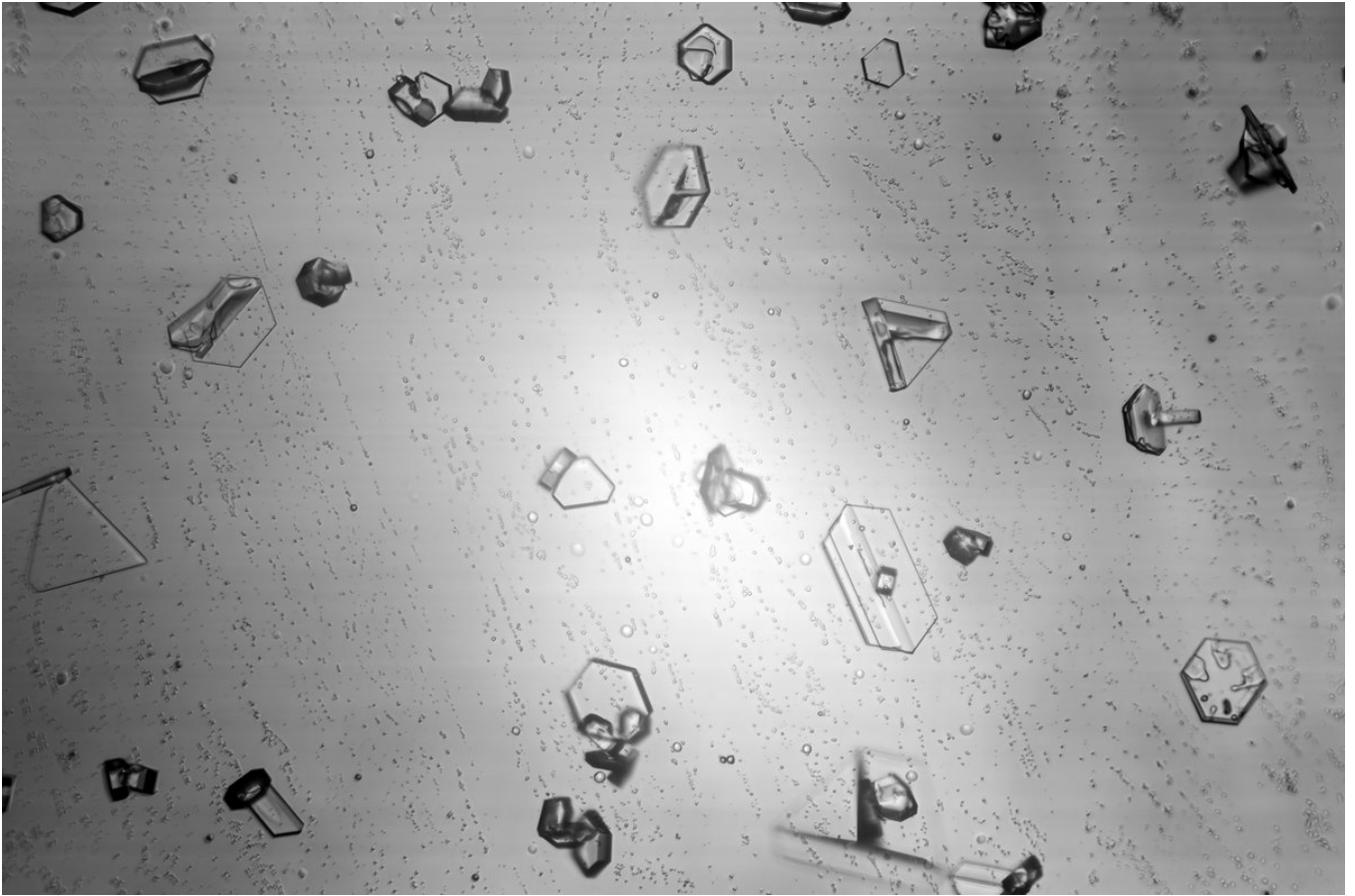
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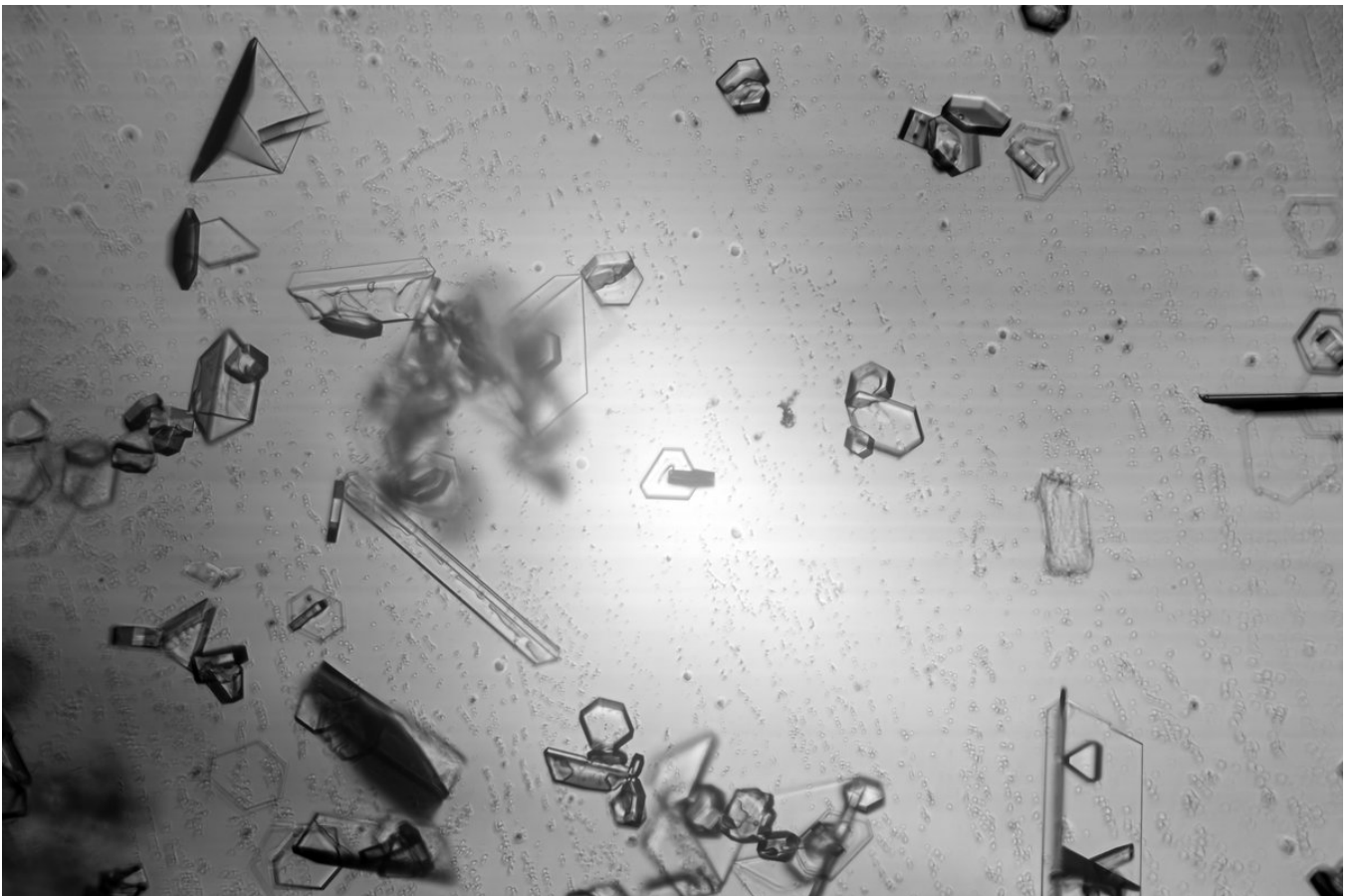
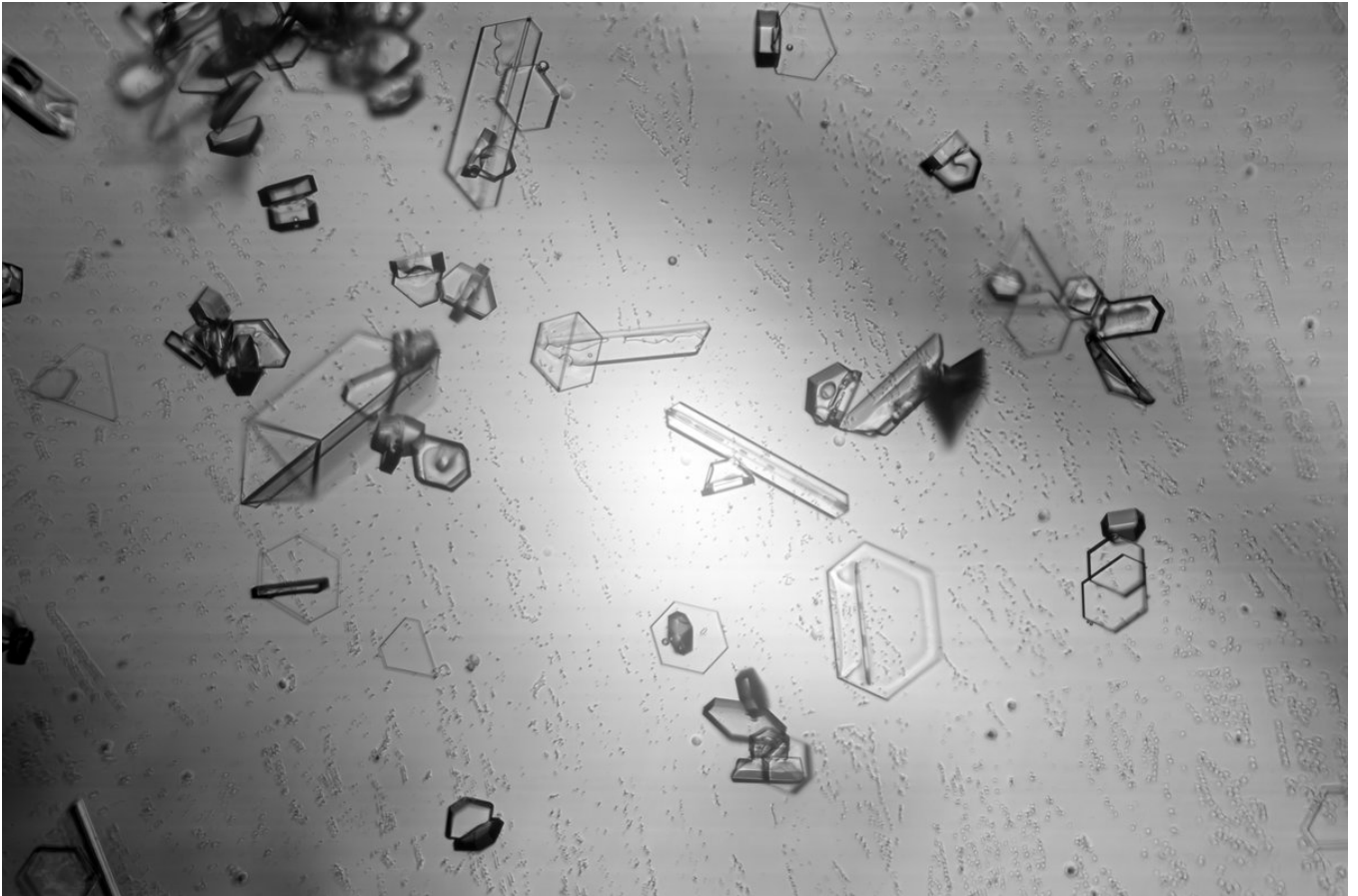




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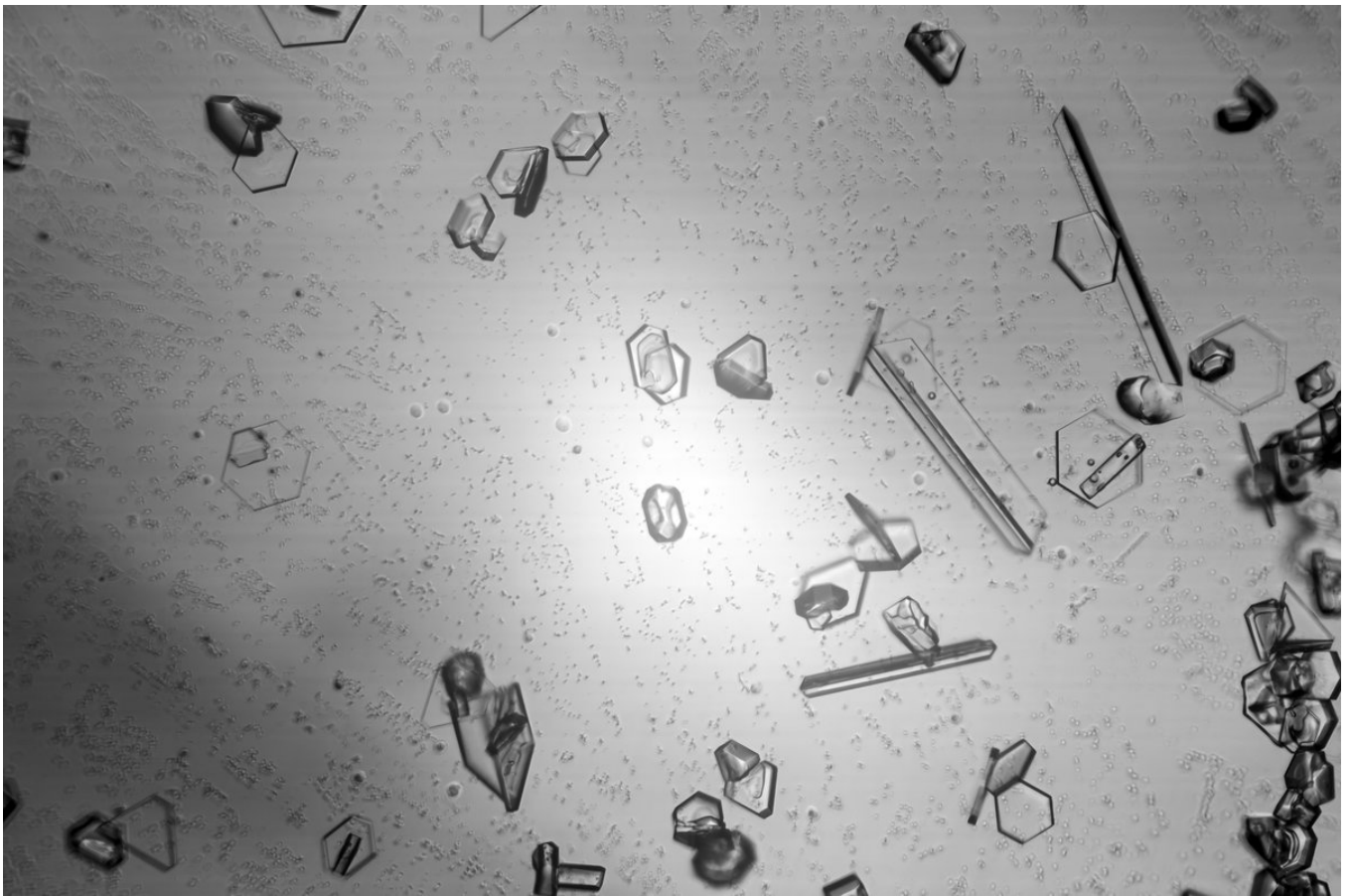
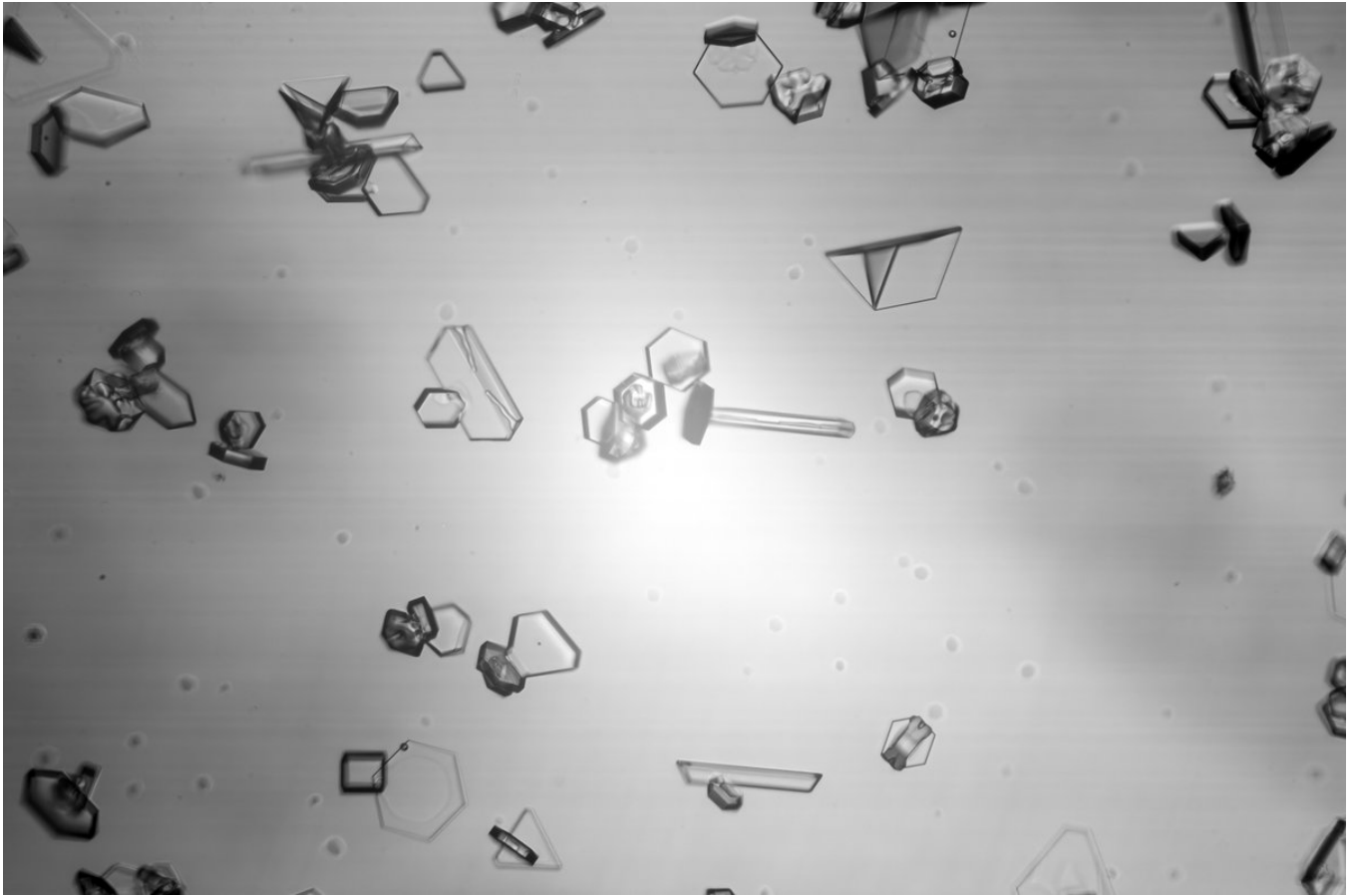
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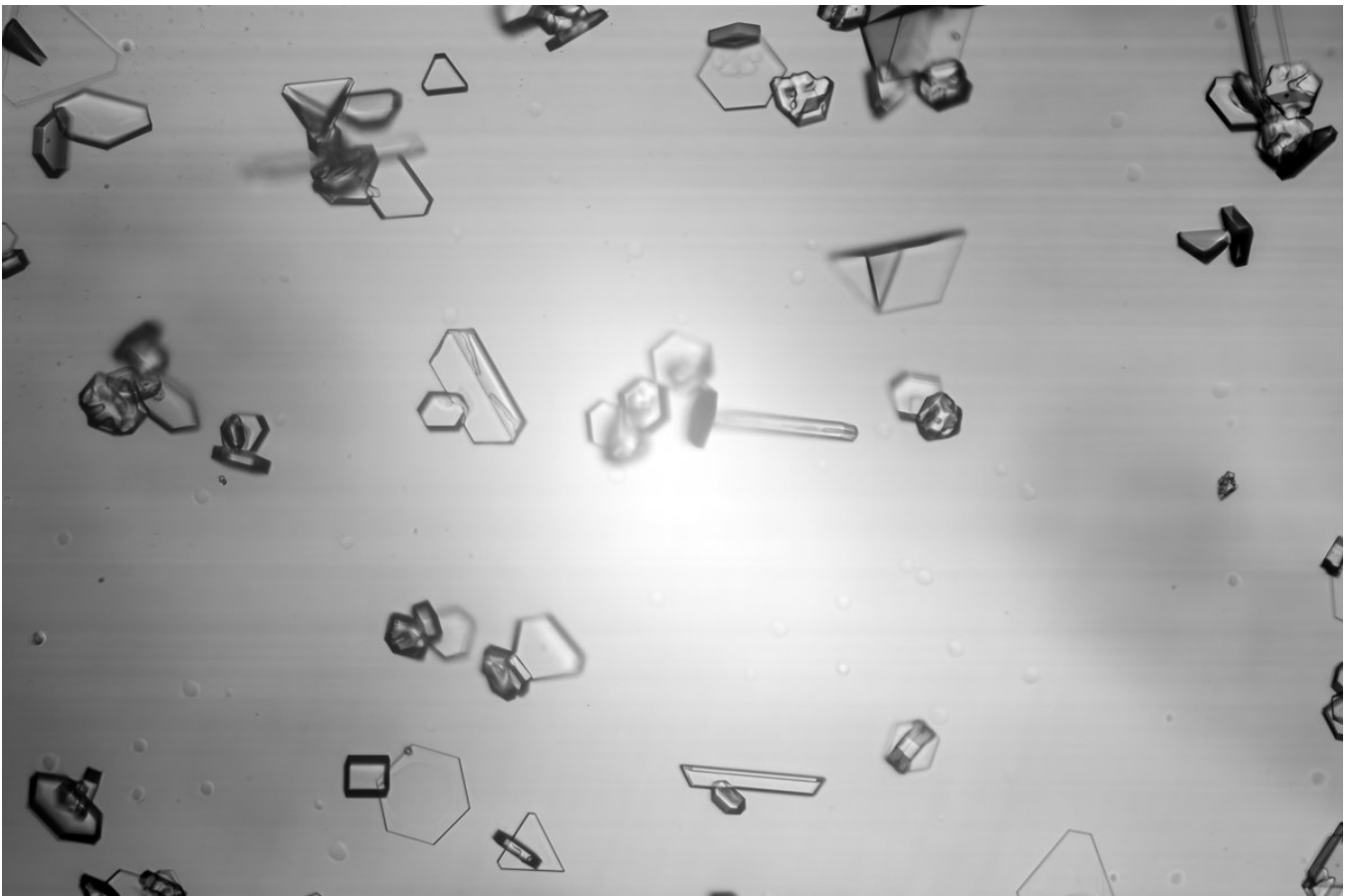
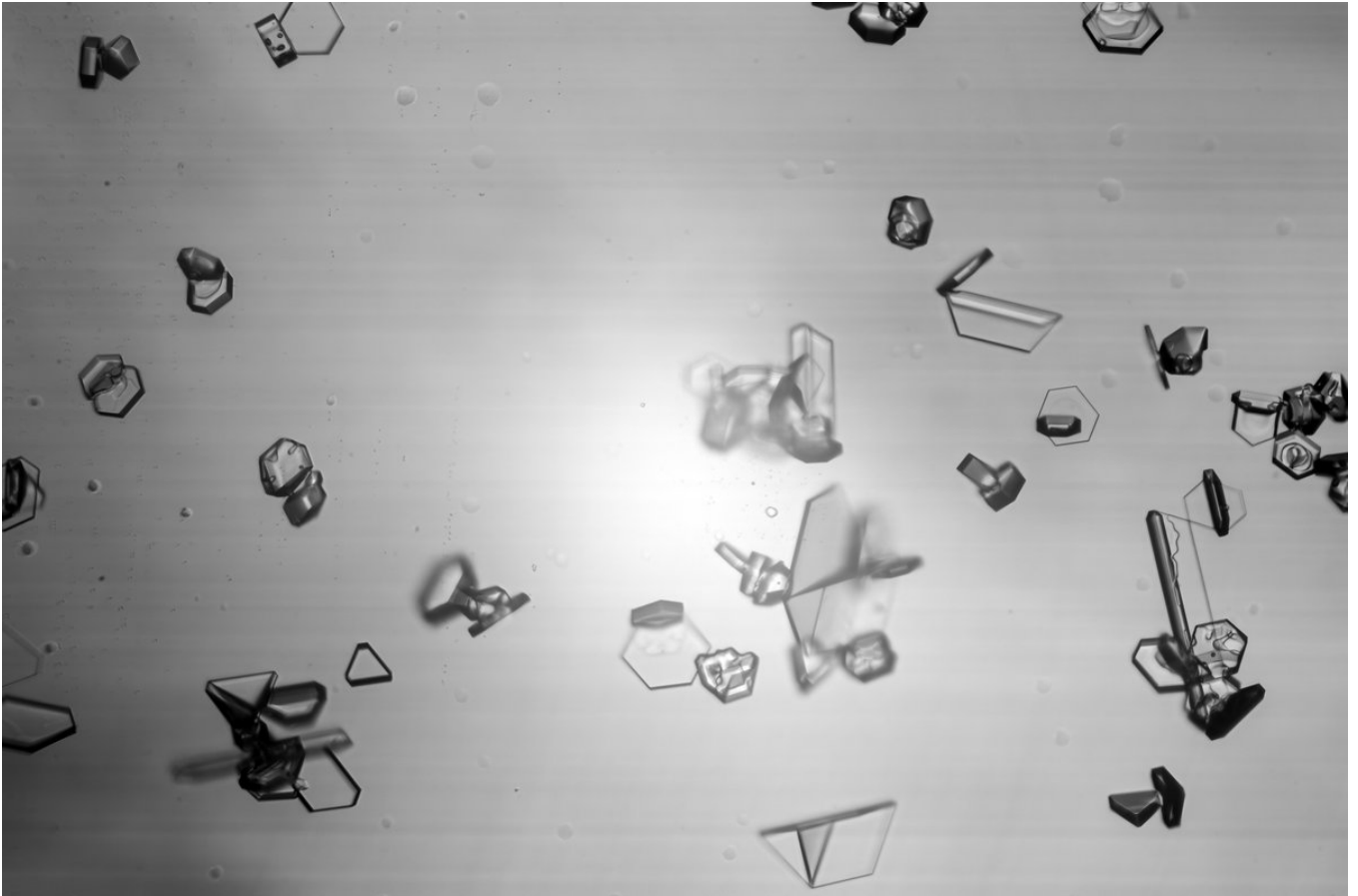




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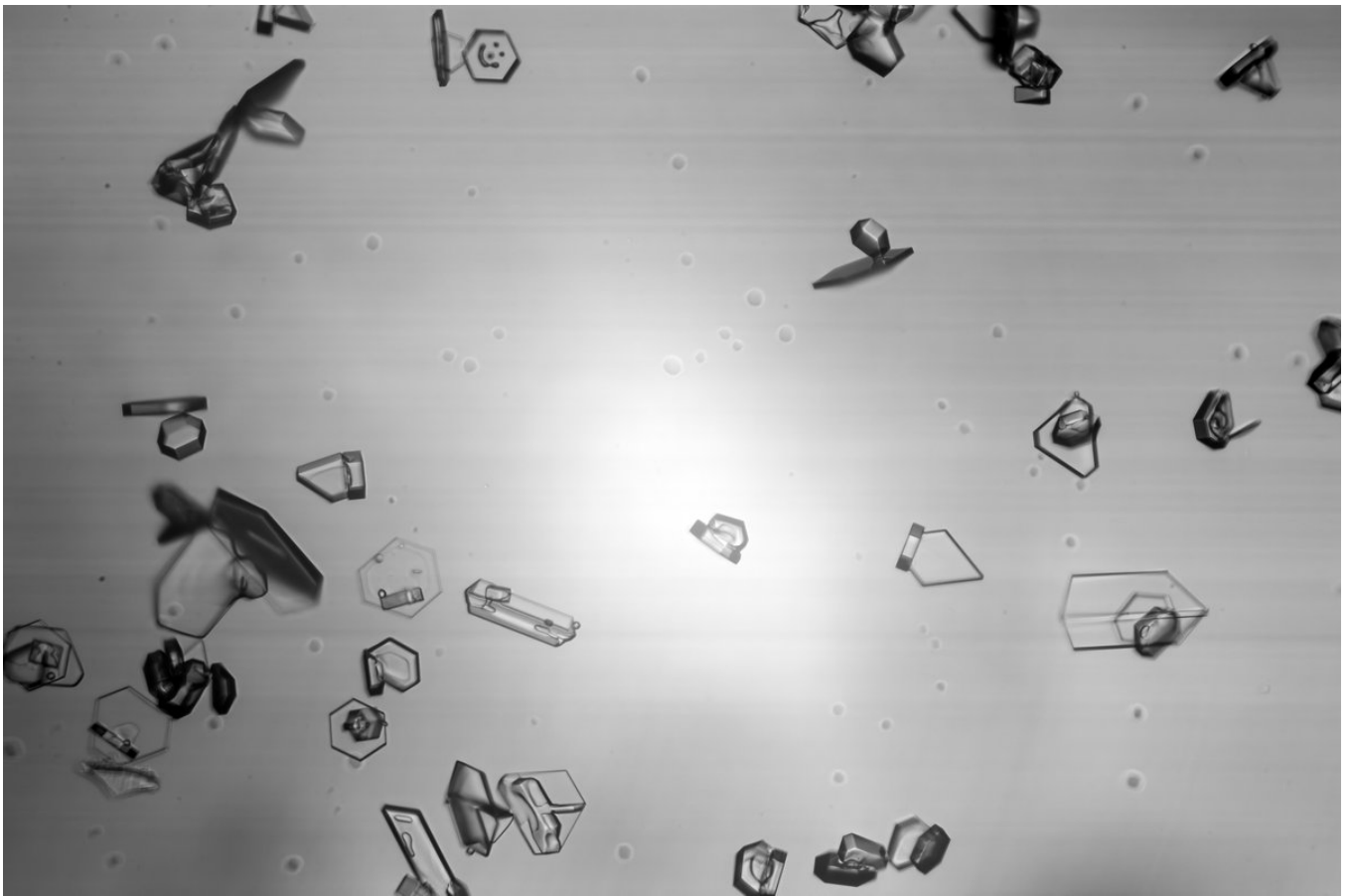
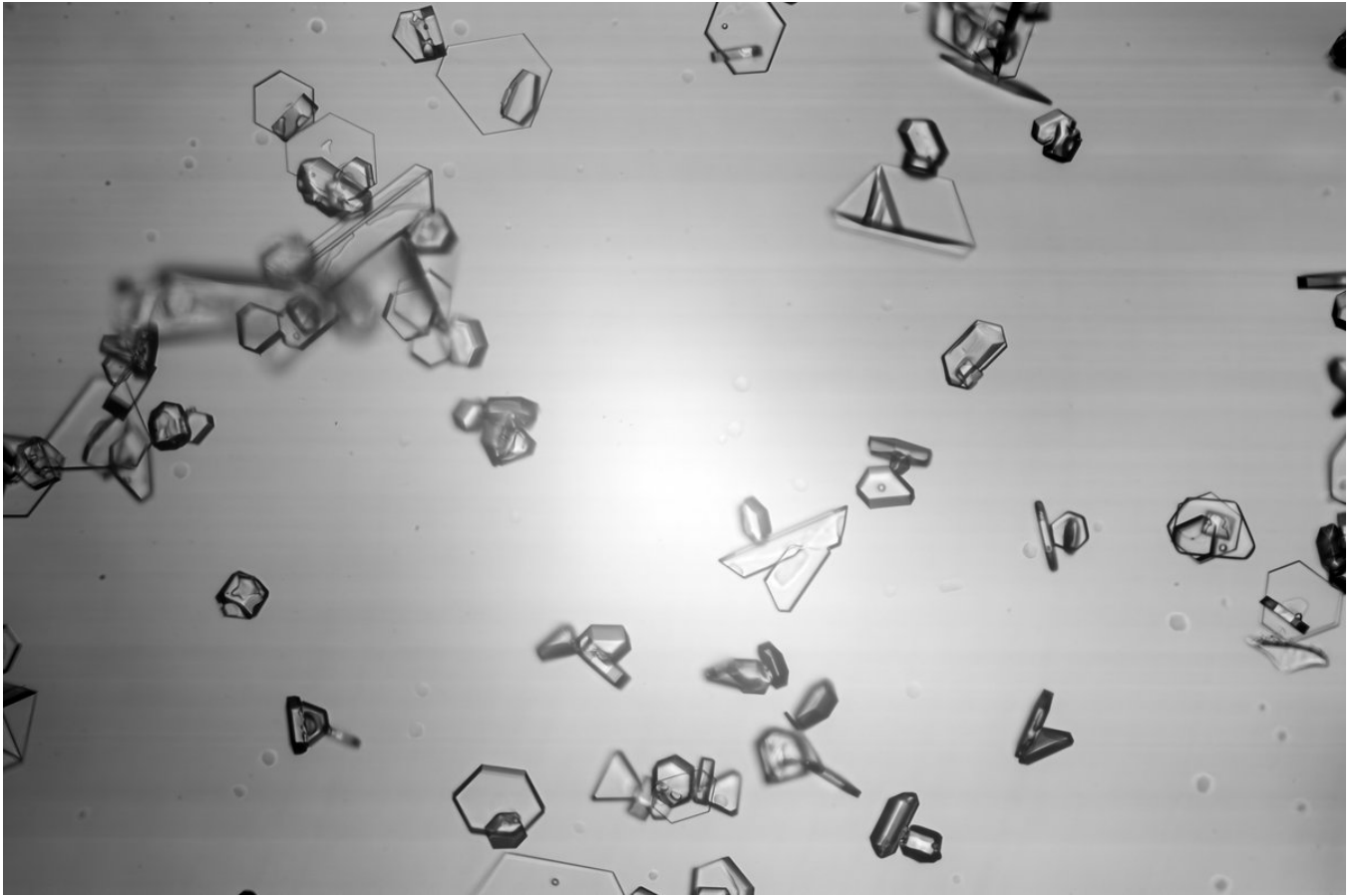
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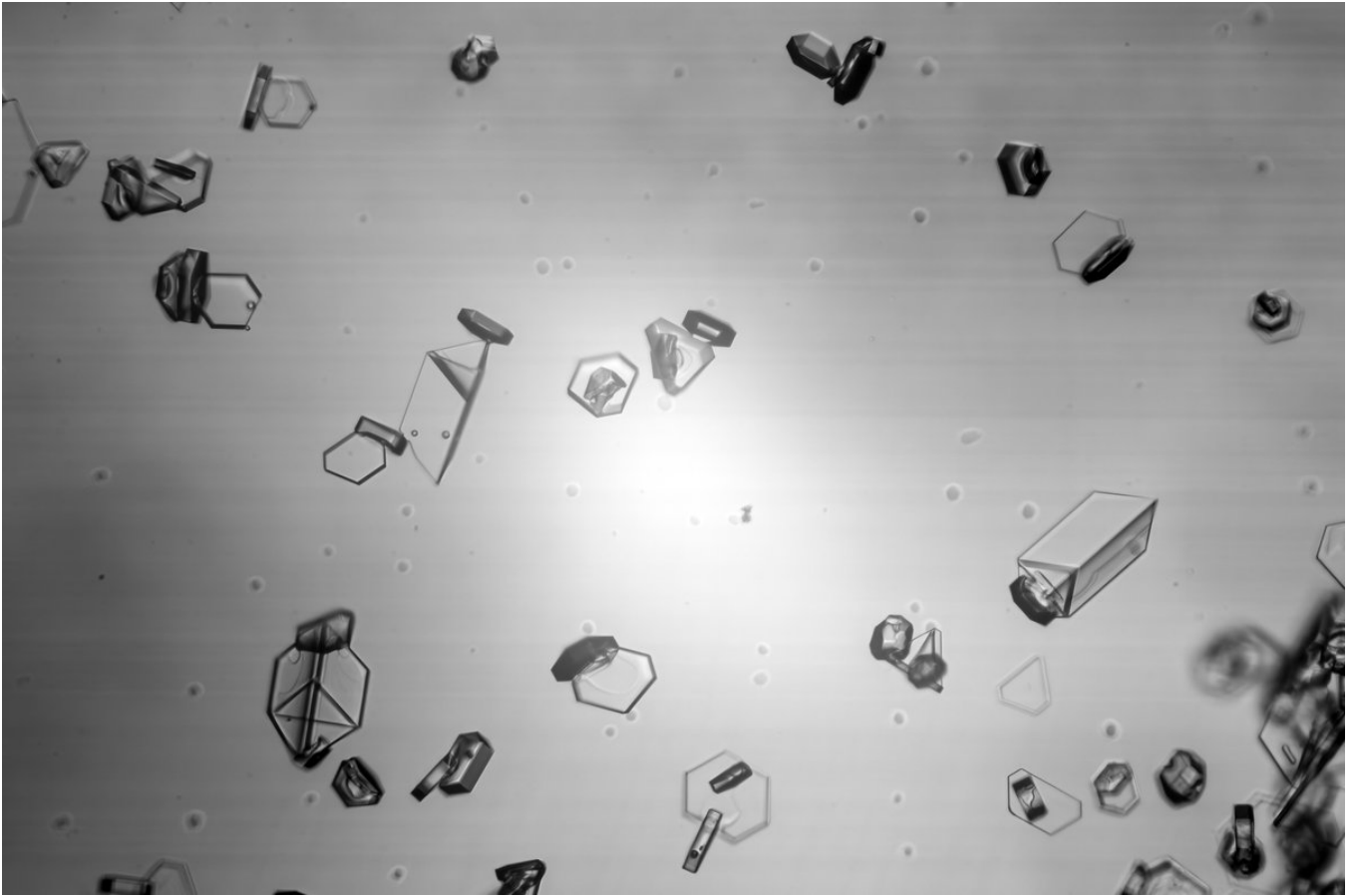




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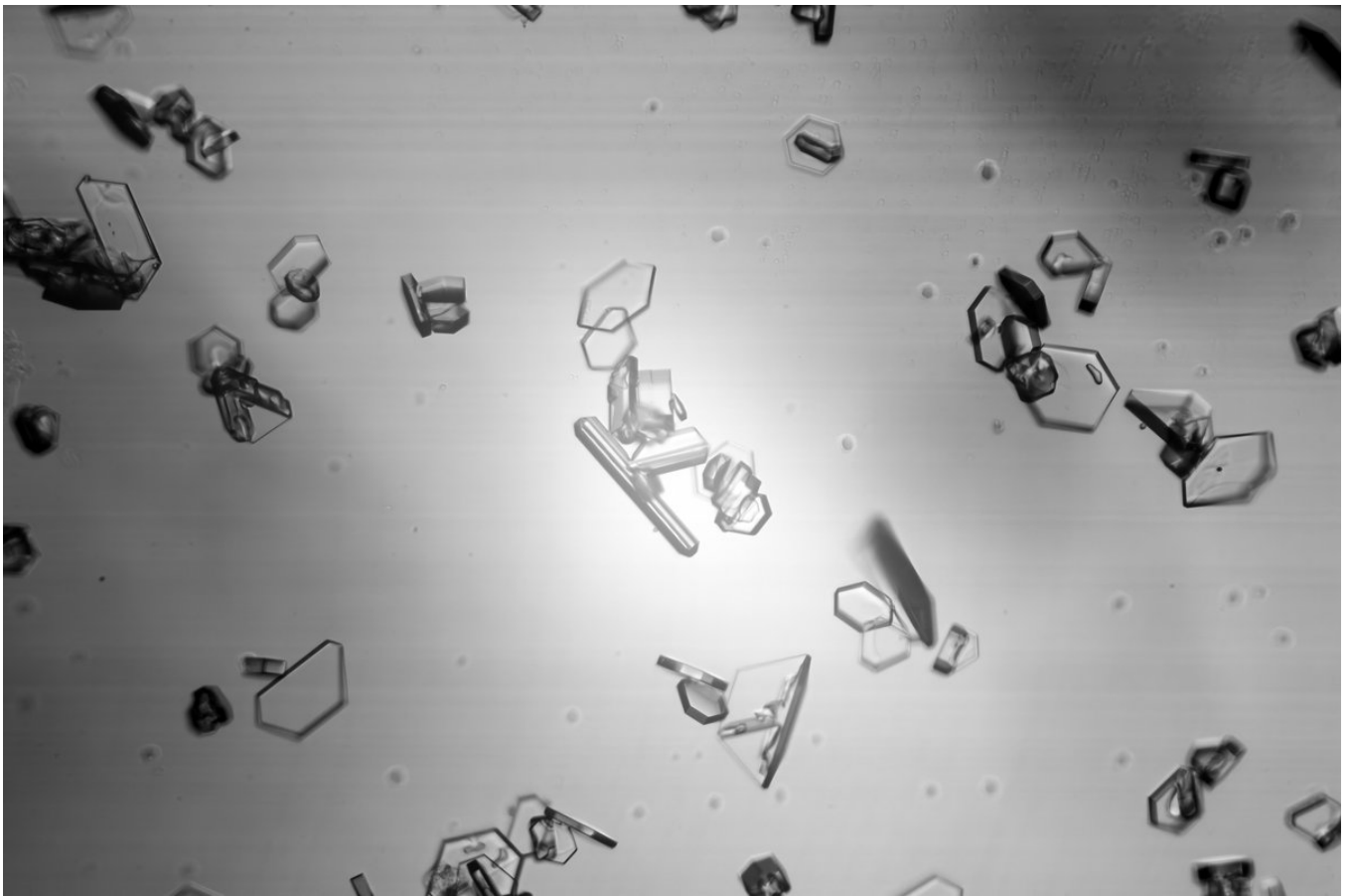
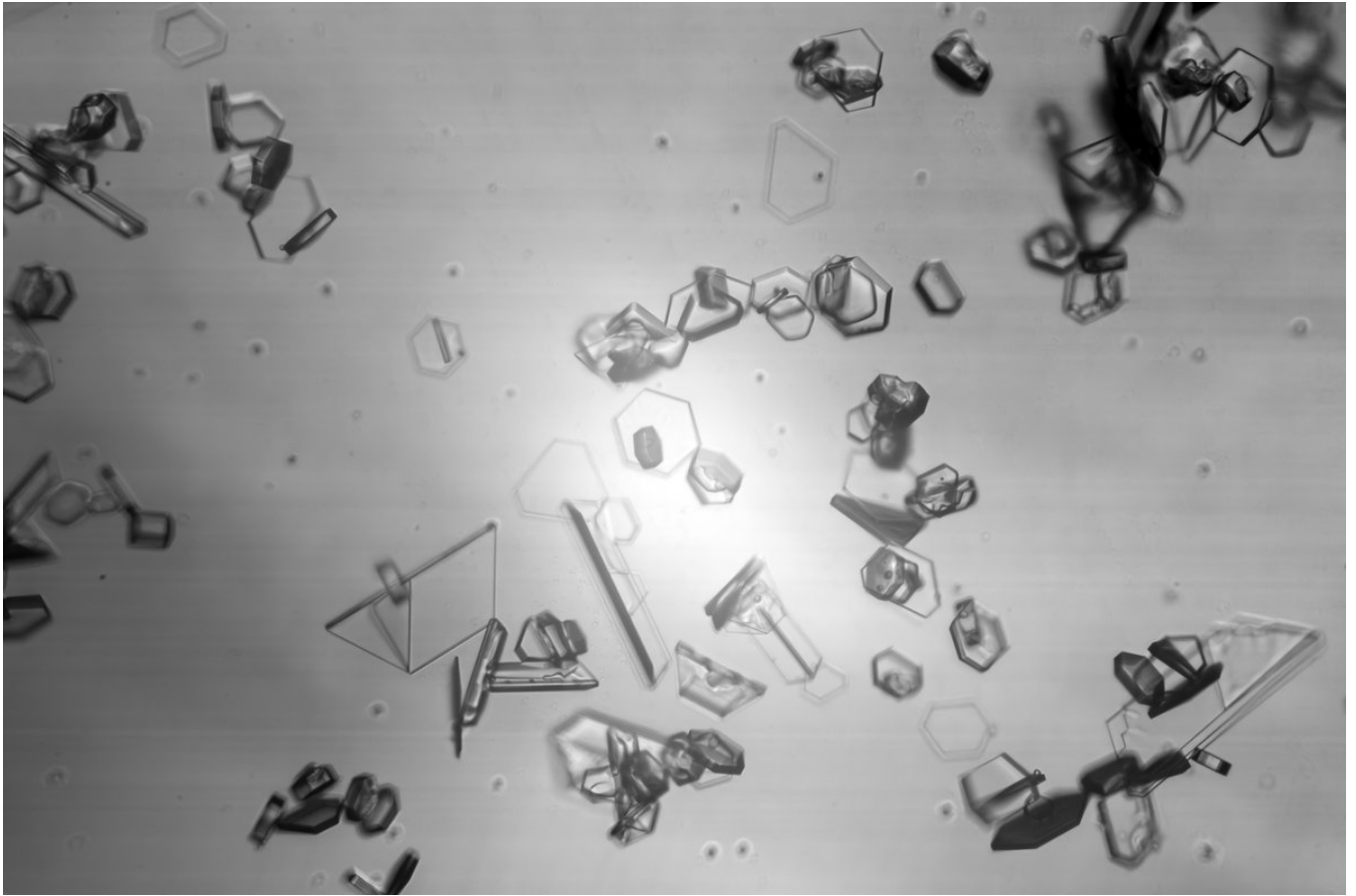
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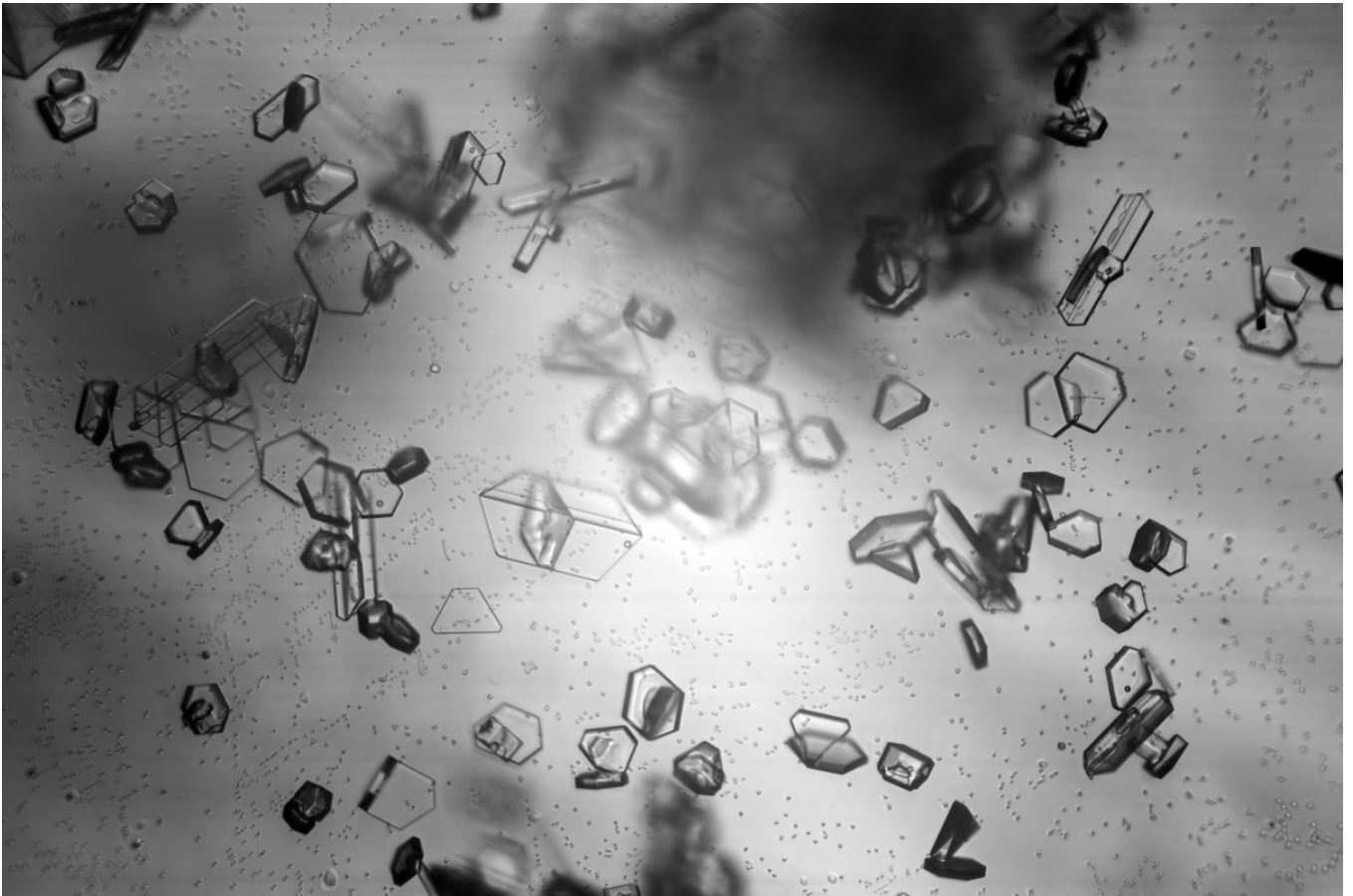
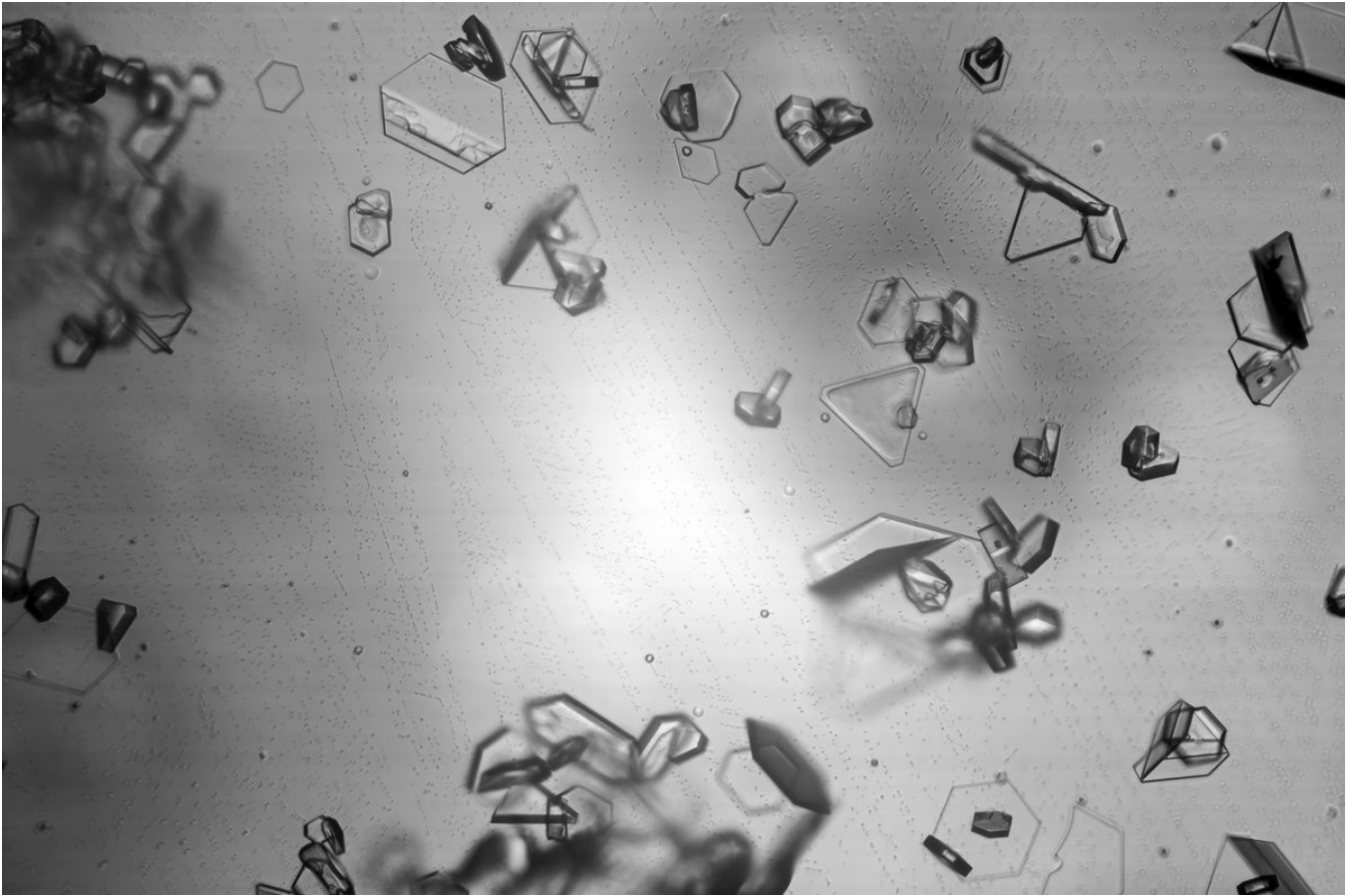




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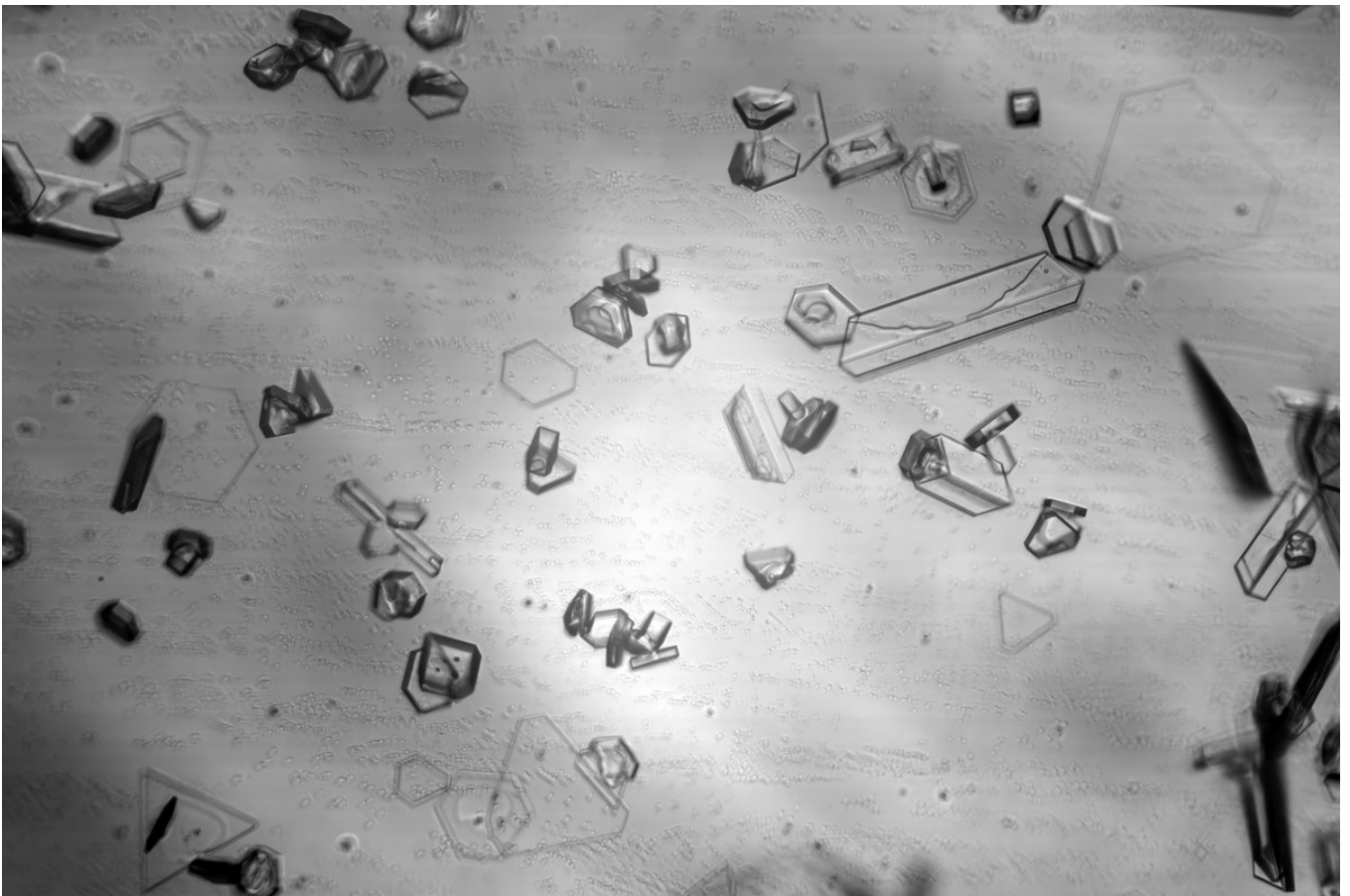
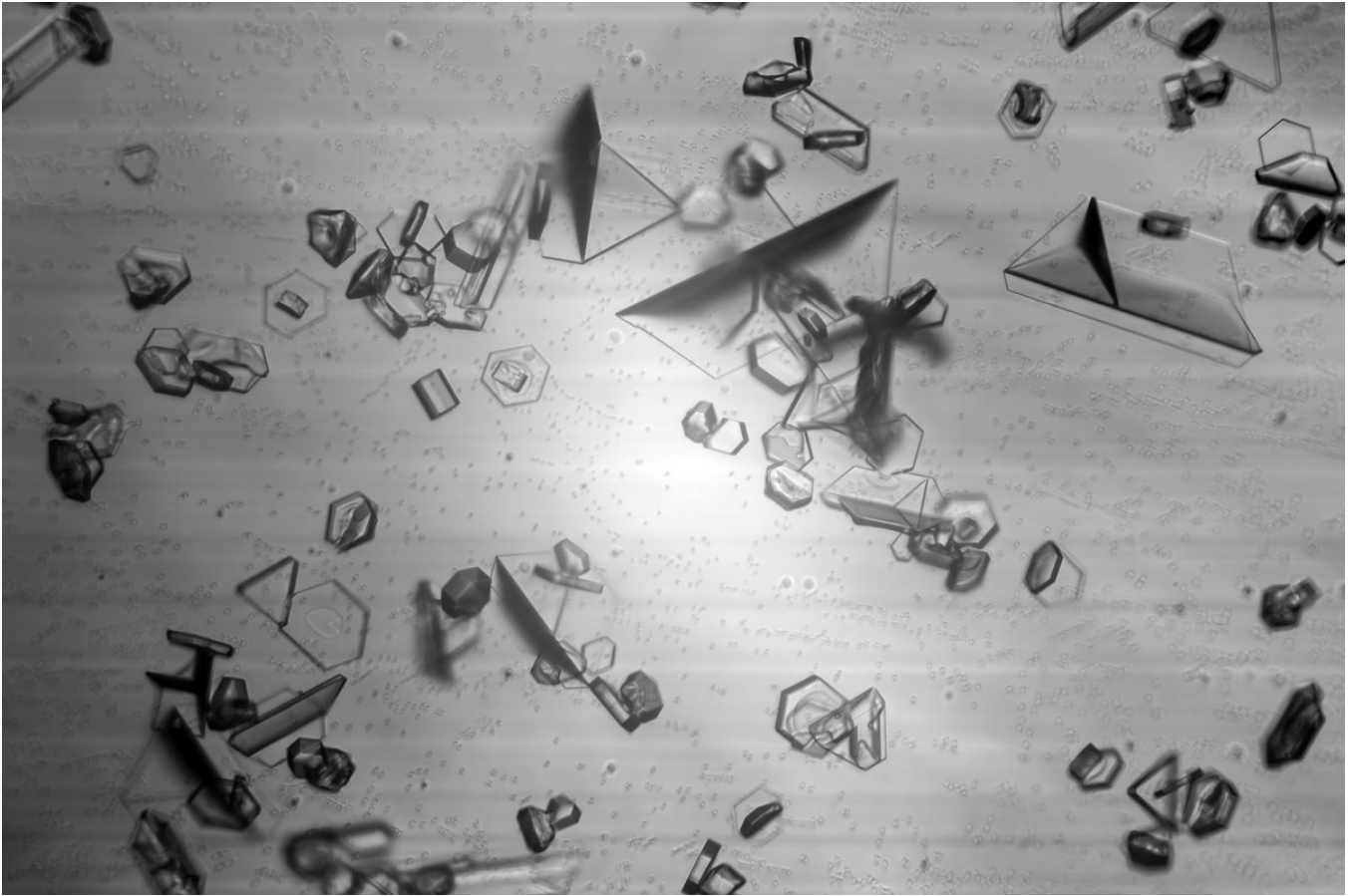
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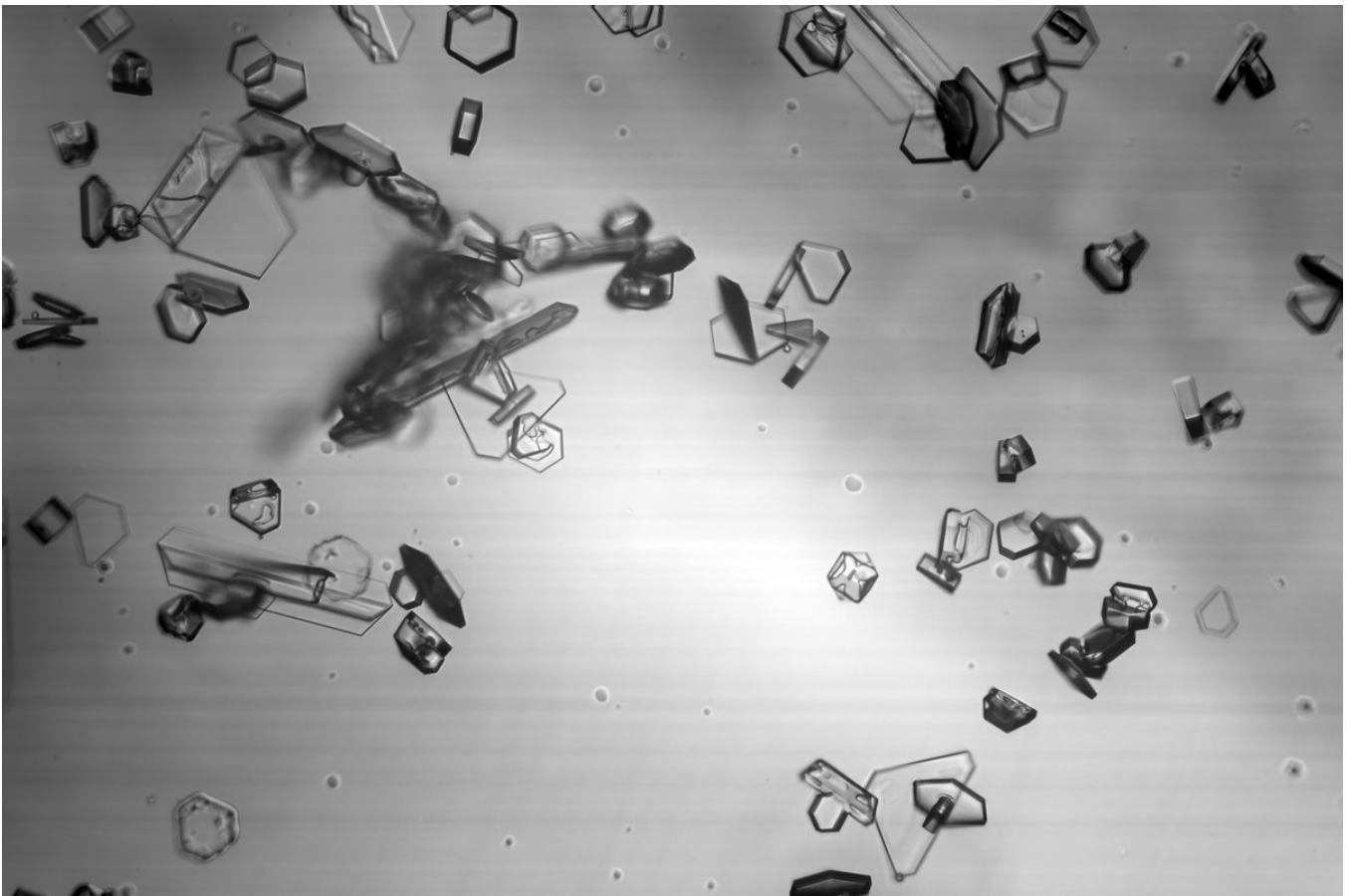
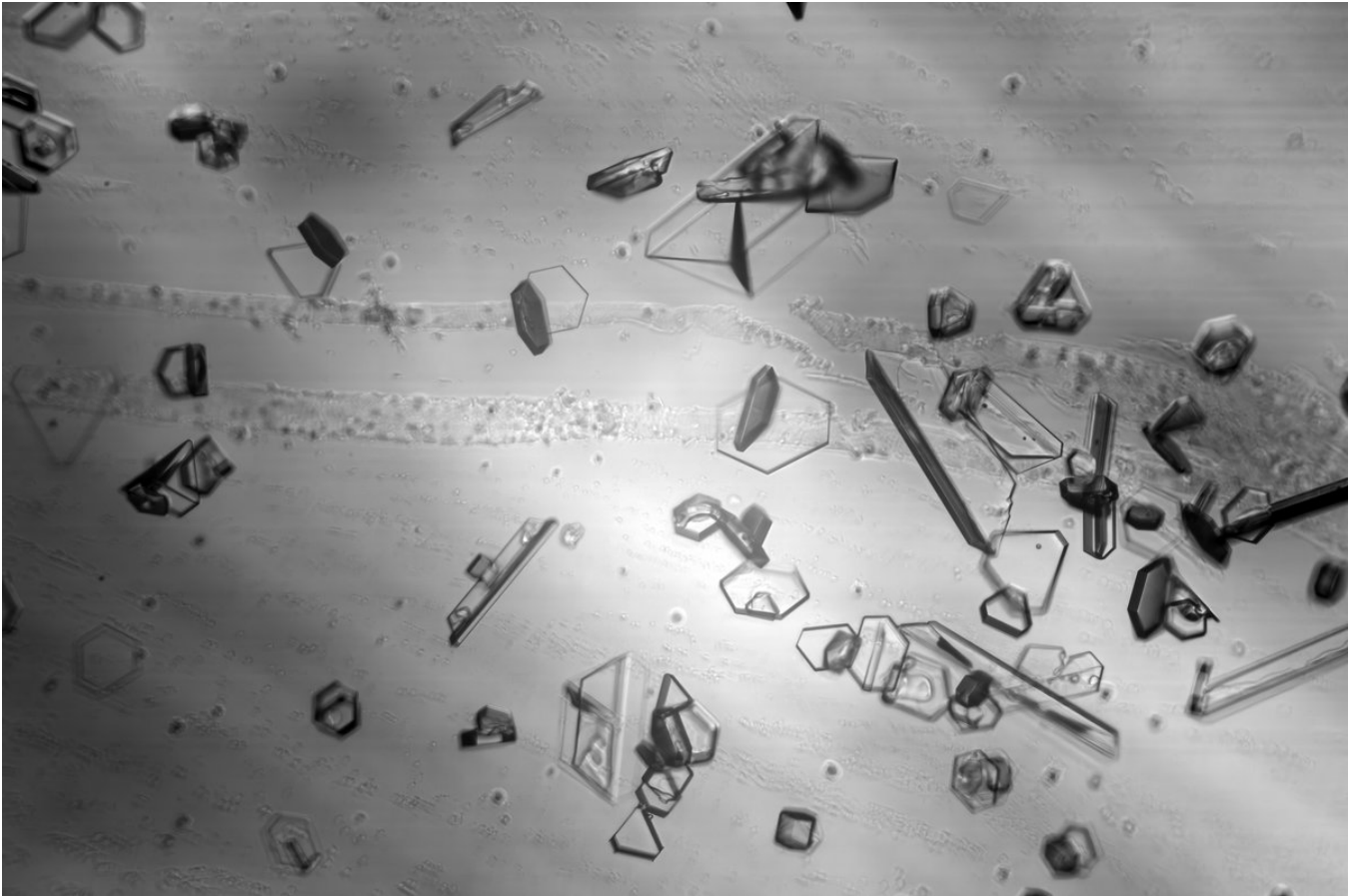




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